

Measurement of Crowding in Embedding Spaces

Sunil Pedapudi*

July 2026

Abstract

Embedding-based systems — retrieval, retrieval-augmented generation, deduplication, clustering — depend on a single geometric property: items that must be distinguished must maintain distinction while conforming to a similarity metric. We call the failure of this property *crowding*. Crowding is a local phenomenon: a handful of documents can collapse together within an embedding space while the global view of the corpus appears issue-free. This report describes *ambit*, an instrument that measures how a corpus occupies its embedding space. It treats an embedded corpus as a pattern of points on a sphere and applies techniques from various scientific fields related to point patterns: are some points packed more tightly than chance would produce, how tightly, and which ones?

ambit earns its robustness by measuring distance between embeddings and calibrating against a mathematically ideal distribution of points to detect tightly-packed clusters. It then translates those distance measurements into a statement about retrieval. Because a query never lands exactly on its target, when two documents sit close together a query aimed at one can return the other — a *collision*. Working only from the distances between documents, with no labels or example queries, *ambit* computes how many collisions to expect at any level of query imprecision. The largest imprecision a corpus absorbs before collisions begin is its *resolution bandwidth* σ^* .

ambit is compared to other measurement techniques to quantify properties of embeddings: mean cosine, isotropy, and hubness. On synthetic corpora built with no crowding, *ambit* has low false positive rate. When near-duplicate groups are inserted into a corpus, *ambit* recovers them in full; mean cosine, isotropy, and hubness miss them. *ambit* was also applied to five public collections embedded by seven encoders. Its per-document measurements, computed before any relevance labels were seen, predicted which documents retrieval would go on to fail, and the predictions held on every collection except multi-hop question answering — where failure comes from needing a second document rather than from crowding.

On a million-document legal corpus, the repairs *ambit* chose — merging the near-duplicate documents it names — improved retrieval without harming any query. Added as penalty terms to a supervised fine-tune, the same measurements halved the damage that fine-tune caused outside its training data at no cost to its gains, and a label-free rule built on them selected the better of two fine-tuned models before any label was read.

*The *ambit* project. Code and reports: <https://github.com/pedapudi/ambit>.

1 Introduction

An embedding model maps items to vectors so that semantic relatedness becomes geometric proximity. Once a corpus is embedded — typically onto the unit sphere $\mathbb{S}^{d-1}$, with similarity as the cosine of the angle between items — every downstream capability reduces to one primitive: *given a direction, list the items near it*. The primitive degrades in a specific way when distinct items land too close together: queries cannot separate them, and the system exhibits impostors at the top of rankings, near-duplicates that cannot be ordered, and clusters that blur into one another. We refer to this failure mode as *crowding* and to its complement — the corpus’s ability to keep distinct items distinct — as *resolution*.

An embedded corpus is a *pattern of points on a sphere*, and crowding is a question about the arrangement of those points: do some of them bunch together more tightly than chance would produce, how tightly, and which points are bunched together? Other sciences ask the same questions of their own point patterns: ecologists of trees in a forest plot, epidemiologists of disease cases, astronomers of galaxies. Half a century of developing statistical methodology has yielded techniques for answering those questions. The methods they produced have two properties of interest: they check for density of points at every possible distance between points (rather than at discrete distances) and they judge that pattern against an ideal model of no clustering. Embedding geometry is the same measurement problem, so the techniques can be applied on a new domain: **ambit** applies those methods to the high-dimensional sphere and translates their output into the language of vector search. §3 credits each method’s source.

Seen this way, crowding is difficult to measure for two reasons, which §2 and §5 explain further. It is *local*: a collapsing pocket barely moves any global metric. And the standard instruments for localizing density — histograms, hexbin lattices, voxel grids, fixed- k neighbor counts — carry hidden discretization parameters whose values change the answer. Figure 1 illustrates both difficulties.

This report describes **ambit**, an instrument built to measure crowding without relying on global averages or on a discretization. There are three design principles shaping **ambit**. **ambit** measures crowding at every similarity level — every value that cosine similarity can take — rather than at one chosen cutoff, to avoid discretization (§5). It reads each measurement against a model of what the corpus would look like with no crowding — a *null model* — so that the mild concentration that encoders are trained to produce is not mistaken for a defect (§2). And it names individual documents so the measurements can be applied in practical settings related to the specific crowded documents. §2–§8 build and validate the instrument under these commitments; §10 and §11 use its measurements to route deduplication and to guard a supervised fine-tune. **ambit**’s contributions are defined as what it adds beyond the prior work it builds on (§3):

1. **A calibrated instrument for measuring crowding** (§6, §9). A point-pattern statistical toolkit applied to embedding spaces. It returns four readouts:
 - the *crowding curve* — for every similarity level, the fraction of document pairs at least that similar;
 - an *occupancy discrepancy* — one number for how unevenly the corpus fills the space. It condenses the whole curve, so a corpus can be tracked across encoder

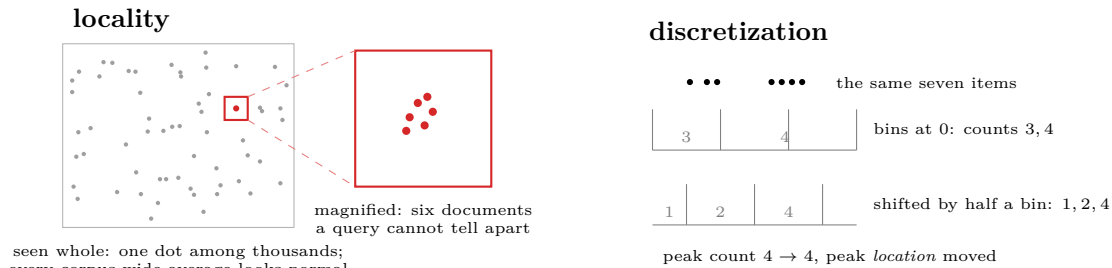


Figure 1: The two measurement difficulties visualized. Left: viewed whole, the corpus shows nothing unusual — the defect is one dot among thousands, and on the benchmark of §4, inserting 200 near-duplicates into 4,000 items moves mean pairwise similarity by only ≈ 0.002 on a scale of 1 (§2.3). Magnified, that dot is six documents a query cannot tell apart, each expecting 2.6 retrieval collisions against zero for a well-spread corpus. The pathology is severe, local, and invisible to any average. Right: binning the same seven items with the same bin width but a half-bin origin shift changes the counts and relocates the peak — the discretization, not the data, chose the answer. §5 quantifies this on the tool’s own displays.

changes, repairs, or drift without comparing curves by eye;

- a *per-document crowding radius* — how far a document must reach to find company, which provably moves only slightly when the corpus changes slightly;
- a *catalogue of tight groups*, found without choosing any similarity cutoff.

Each is read against a simulated corpus of the same size with no crowding. One such reference reproduces the narrow angular region into which learned encoders pack every document, so that this benign effect is not counted as a defect.

2. **Retrieval semantics for crowding measurements (§7).** An exact conversion from the distance between two documents to the chance that an imprecise query aimed at one returns the other. Summed over a corpus it gives the resolution bandwidth σ^* : how imprecise a query may be before the corpus starts returning the wrong document. The same conversion also explains a setting practitioners currently choose by convention: the temperature of the widely used uniformity training loss [7] corresponds to a specific level of query imprecision. This bridge from crowding statistics to retrieval error counts is, to our knowledge, new.
3. **Experimental evidence that geometry predicts retrieval failure (§11).** We measured five public collections under seven public encoders and locked every readout before opening the answer keys. The per-document score then identified the documents that retrieval would later fail on, in 26 of the 29 pairings that test it (sign test $p \approx 8 \times 10^{-6}$). All three exceptions are multi-hop questions, where failure comes from needing a second document rather than from crowding. The corpus-level σ^* ranks encoders only where the task is similarity matching. Both boundaries come from measurements rather than assumptions.
4. **Measurement as a governor for fine-tuning (§10).** ambit’s measurements can select a data repair, and can then confirm that the repair worked, without labels at

either step. On the case-study corpus, merging the near-duplicate documents the measurements had named, at the similarity they identified as true duplication, improved twelve queries and harmed none ($p = 0.0005$); the same merge at a looser similarity made retrieval worse. Added to a supervised fine-tune as penalty terms, the measurements halved the damage that fine-tune caused outside its training data at no cost to its gains, and the terms provably act only on pairs close enough to be confused. Used as the training objective itself rather than as a supplement, the same measurements leave retrieval unchanged (§10.4).

Throughout, **ambit** is *unsupervised by design*: no labels, gold pairs, or relevance judgments are consumed. The mathematics assembled here is, with small exceptions, classical, drawn from spatial statistics, directional statistics, geometric inference, and discrepancy theory (§3). The contribution is the *instrument*: the selection, calibration, and combination of these known statistics into a measurement system with well-understood properties.

2 Geometry of Crowding

Three elementary observations about high-dimensional spheres determine the measurement design.

2.1 Items are directions; a retrieval neighborhood is a patch of the sphere

Embeddings compared by cosine live on the unit sphere: each item is a *direction*, and similarity is the angle between directions. A retrieval neighborhood — “everything scoring above 0.8 against this query” — is the set of directions within a fixed angle of the query, which is a patch of the sphere’s surface bounded by a circle (a *spherical cap*; Figure 2). Every question about how the corpus fills the space then becomes a statement about caps. A corpus is concentrated when some caps hold more items than their share of the sphere’s area predicts. An item is crowded when even a small cap centered on that item already contains other items. And two items are confusable when they share one small cap: any query that lands near one cannot avoid the other, so they are *future collisions*.

2.2 In high dimension, “random” means “orthogonal”

Fix one item on $\mathbb{S}^{d-1}$ and picture the set of directions at least 0.3-similar to it: a cap around it (Figure 2). In low dimension that cap is a sizeable patch of the sphere. As dimension grows, there are ever more ways for two directions to differ, so the sphere’s area migrates toward the thin band 90° from *any* fixed direction. The cap’s share of the surface collapses: at $d = 768$, the cap at $\cos \geq 0.3$ holds roughly 10^{-16} of the sphere’s area. A uniformly placed second item essentially never lands in it. Numerically, the cosine between two independent uniform directions follows a bell curve centered on 0 whose width shrinks as $1/\sqrt{d}$ ($\mathcal{N}(0, 1/d)$). At $d = 768$ that width is $1/\sqrt{768} = 0.0361$, matching the 0.0361 measured on 4,000 sampled uniform points (Figure 3). Three consequences follow.

- **“Similar” is a geometrically tiny target.** Two unrelated items land in each other’s $\cos = 0.3$ caps about once in 10^{16} draws.

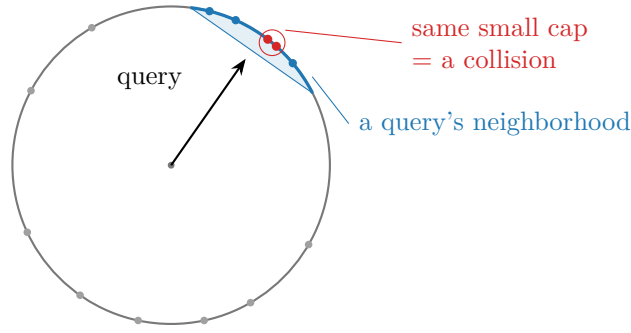


Figure 2: An illustrative example. Items are directions on the unit sphere, and a retrieval neighborhood is a cap. The two red items share one small cap, so no query landing there can tell them apart: the geometric definition of a retrieval collision.

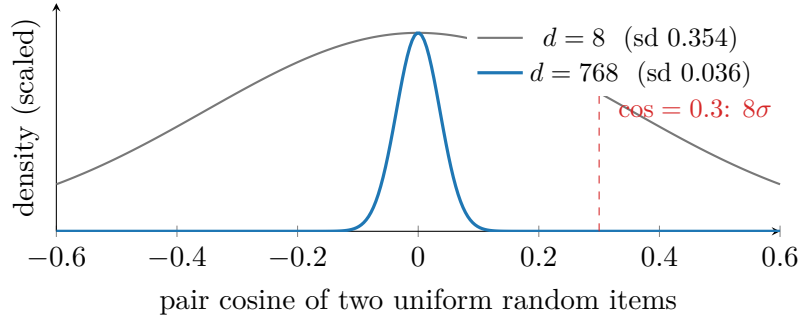


Figure 3: Concentration of measure: the null pair-cosine distribution collapses to a narrow spike at 0 as dimension grows. The informative region of the null is empty, and every average is dominated by that spike. Densities are analytic; the $d = 768$ width is the *measured* sd 0.0361 from 4,000 sampled uniform points, matching $1/\sqrt{d}$ to four decimals.

- **The well-spread null has no small-distance tail at all.** In each of 19 simulated uniform corpora of this size, the fraction of pairs with $\cos \geq 0.30$ is exactly zero, so *any* observed mass there is unambiguous structure. (Nineteen is the classical envelope count: real data exceeding all 19 simulations is a $1/20 = 5\%$ event under the null.) Distance concentration — the classical curse of high dimension [40] — is therefore an asset here: every close pair is a finding.
- **Averages over pairs are dominated by the background**, since almost every pair is a near-orthogonal background pair. The next subsection makes this precise.

2.3 Why averages cannot see crowding

Every corpus-wide average of a pairwise quantity lives on the *square of all pairs*: an $n \times n$ grid with one cell per ordered pair of items (Figure 4). A group of crowded items occupies a stretch of the square’s *side*; its pairs occupy the small square block where that stretch meets itself. Side and area scale differently: membership grows with the side, while the group’s *pair share* — the fraction of all pairs that fall inside it — grows with the area. A pocket

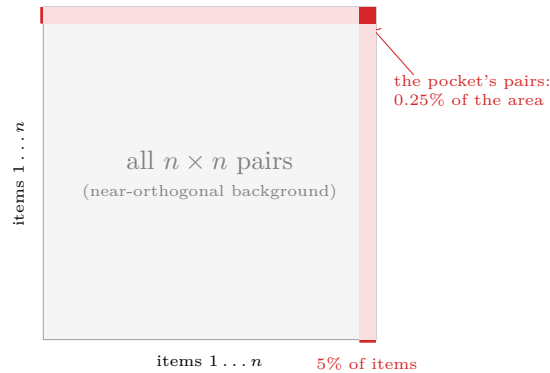


Figure 4: The square of all pairs (schematic; percentages from the $d=768$ benchmark). Every average of a pairwise quantity is taken over this square’s area. Crowded items occupy a stretch of the *side*; their mutual pairs occupy only the small block where that stretch meets itself — 5% of the items, 0.25% of the pairs. The pale strips are the pocket’s pairs with the bulk, which are near-orthogonal like the background. Averages weight by area, so a catastrophic pocket is geometrically diluted before any arithmetic begins.

holding 5% of the items therefore owns only 0.25% of the pairs any average is taken over. The benchmark corpus used throughout this report makes the dilution concrete. It holds $n = 4,000$ items at $d = 768$, and the last $k = 200$ of them — five percent — are replaced with near-duplicates of one another, so alike (median pairwise cosine 0.929) that a retrieval system cannot rank them apart. §4 gives the full construction. Now follow the arithmetic of the average, step by step. Five percent of the items yields $0.05^2 = 0.0025$ of the pairs, so 99.75% of the average consists of background pairs, which the previous subsection showed sit near $\cos = 0$. Each pocket pair contributes a cosine of roughly 0.93 at that 0.25% pair share, so the entire catastrophe adds $0.0025 \times 0.93 \approx 0.00023$ to the mean. Two hundred documents became interchangeable, and the corpus-wide average moved from about 0.000 to 0.002: invisible without a calibrated test, and uninformative about *which* items failed. Prior art such as covariance-spectrum summaries (effective rank, isotropy scores) fails for the same arithmetic reason: they are sums over the corpus, and sums are dominated by the diffuse majority. Global averages address whether the whole cloud is degenerate, while crowding concerns which part of the cloud has collapsed, how tightly — at $\cos = 0.9$, say, rather than at $\cos = 0.5$ — and which items it contains. A sound instrument must therefore do four things, and §6 builds each of them.

- **Read whole distributions rather than averages.**
- **Read them at every similarity level**, since pockets live at the tight levels the bulk never reaches.
- **Calibrate against a matched null**, since an absolute cosine value carries no meaning on its own.
- **Name the individual items**, since the actionable output is a list of ids.

3 Related Work

The introduction framed an embedded corpus as a point pattern and crowding as the clustering question the observational sciences ask of point patterns. Those sciences have mature statistics for answering it, and **ambit** applies them to the high-dimensional sphere where embedded corpora live. Four fields contribute:

- **spatial statistics** — the crowding curve and the test that decides whether it is significant;
- **geometric inference** — the per-document crowding radius and the catalogue of tight groups;
- **discrepancy theory** — the occupancy discrepancy;
- **directional statistics** — the reference that reproduces the cone.

Each paragraph below first describes a method as its home field uses it, then states what changes on embeddings. None of these fields supplies the retrieval layer: none has a notion of one document outranking another at a query. The conversion of pair distances into expected retrieval collisions (§7) is therefore **ambit**’s own, and it is what turns these borrowed statistics into a retrieval instrument.

Counting pairs at every distance. Ripley’s K function [16, 17] is the standard test for whether a set of points clusters. An ecologist applies it to trees in a forest plot, an epidemiologist to the home addresses of disease cases, an astronomer to galaxies in a survey volume, a cell biologist to proteins imaged on a membrane. The recipe is the same in each: count how many pairs fall within every distance, and compare those counts to what scattering the same number of points at random would give. The comparison must be read at every distance, because a pattern can be clumped at one distance and evenly spread at another: trees cluster into groves while the groves sit far apart. Any single radius reports one of these and hides the other.

ambit’s crowding curve is this idea applied to the sphere, and two things change in the move. First, the random-scatter comparison stays exact rather than becoming an approximation: on a sphere the chance that a random point falls within a given distance of another is just the area of the surrounding patch of surface, which has a closed form (the cap-area law). Second, a correction the planar version needs becomes unnecessary. A forest plot has edges, and a tree near one has neighbours only on the inside, so its pair counts run low and the ecologist must compensate for it (an *edge correction*). A sphere has no edge — walk in any direction and the surface continues — so every document sits in the same position and nothing needs correcting [18]. Neither property is a contribution of this report; both are consequences of the domain, and they are noted because they are what makes the transfer clean.

Testing a whole curve at once. Reading the counts at every distance creates a problem of its own. The curve is not one number but a value at each of some eighty similarity levels, and each level carries its own range of counts that crowding-free corpora produce — a band

the observed curve should stay inside. Checking each level separately against its own band does not work: every check carries some chance of a false alarm, and across eighty checks one almost always occurs. On this report’s own crowding-free corpora, that level-by-level reading raised an alarm on all forty (§8).

The repair is to treat the curve as a single object. A global envelope test [21] builds 199 crowding-free corpora, computes each one’s whole curve, and asks where the observed curve ranks among all two hundred by its most extreme departure at any level. That yields one verdict covering every level at once, with a smallest attainable p of $1/200 = 0.005$, since the observed curve can at best exceed all 199 simulations. This is *ambit*’s liftoff test. It also sets a division of labor that the report uses throughout. The band shows at which similarity level a corpus departs from the reference — at $\cos = 0.82$ on the legal corpus, say, rather than anywhere looser. The single global verdict decides whether that departure is real.

Conditioning on an uneven background. Spatial statisticians met *ambit*’s central conundrum long ago. Disease cases cluster in cities, but that says nothing about contagion, because people cluster in cities. Counting pairs against a uniform comparison would measure where the population lives rather than how the disease spreads. The *inhomogeneous K* function [19] repairs this by comparing the pattern against a background that already contains the uneven population, so that only clustering beyond it is reported.

The same repair applies on the sphere. A learned encoder does not spread documents over the whole sphere; it packs them into a narrow region of it, called a cone. That cone is *ambit*’s uneven background. The reference of §6.1 models it with a distribution from directional statistics, the angular central Gaussian, and *ambit* then reports only the clumping that the cone does not explain.

What not to transfer. The same sciences documented the hazards of partitioned statistics. The modifiable areal unit problem (MAUP) [24, 25, 26] and the ecological fallacy [27] are the classical demonstrations that partition-based statistics change with the partition and say nothing about individuals — the two failures, quantified on embedding displays in §5, that *ambit* is built to avoid. The same lesson appears in one dimension as the histogram’s bin-width and origin sensitivity [28, 29, 31]; the averaged shifted histogram [30] repairs placement by averaging over origins, and *ambit* removes the partition altogether.

Auditing every region at once. A *region* here is a patch of the sphere: every direction lying within some angle of a chosen centre. There are infinitely many such patches, at every centre and every size, so asking whether each holds its fair share of documents is an audit that cannot be run directly. Stolarsky’s invariance principle [22, 23] converts it into one computable number, the mean pairwise distance. *ambit*’s occupancy discrepancy is this theorem put to work.

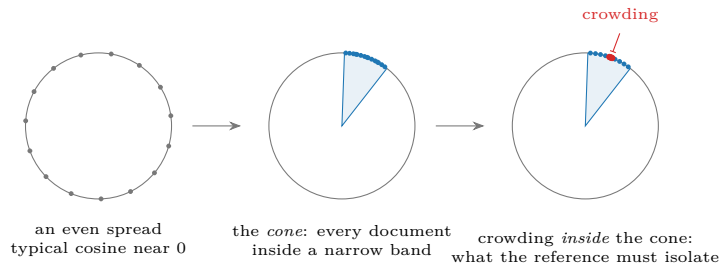
Scoring one point, and finding groups. Two questions remain after the curve: how crowded is *this* document, and which documents form a tight group.

For the first, the *distance to a measure* [32, 33] asks how far a point must reach to gather a fixed share of the data — a small radius means crowded, a large one isolated. It replaces density estimation, which fails in high dimension, and it is stable: changing the corpus a

little provably changes every score only a little. **ambit**’s per-document crowding radius is this estimator on the sphere.

For the second, a *cluster tree* [37, 38] records which groups exist at which density, so groups can be read at all densities at once instead of at one chosen threshold. ToMATo [34] and HDBSCAN’s condensed tree [35] are two standard forms of that object [36], and both inherit stability from persistence theory [39]. **ambit**’s catalogue of tight groups is this machinery with one addition: a group is reported only if it is tighter than the tightest group pure chance produces at the same corpus size.

The cone that every transfer must condition on.



Gao et al. [1] and Ethayarajh [2] established the cone drawn above, and measured its consequence: packing every vector into a narrow band raises all pairwise cosines at once. A corpus can then look crowded everywhere while nothing is harder to tell apart. Judging such a corpus against an even spread would report the cone; judging it against a reference that already contains the cone reports only what the cone does not explain — the third panel. For this reason every readout is calibrated against a reference that contains the cone (an *anisotropy-matched* reference) rather than against the uniform sphere alone. Mu et al. [3] showed that subtracting the mean direction and a few dominant components restores much of the usable angular range. That is the linear repair that **ambit** recommends when its measurements attribute crowding to the cone alone. Cai et al. [4] found genuine cluster structure *inside* the cone, which is what the anisotropy-matched reference is designed to isolate.

What the embedding literature measures instead. Effective rank [5] and IsoScore [6] compress the covariance into one number — roughly, how many directions carry variance, and how round the cloud is. **ambit** reports both as context, and §8 measures that they cannot see small pockets. Hubness [8] — a few items appearing in disproportionately many nearest-neighbor lists — is a symptom of the same congestion that **ambit** measures geometrically, and it serves as a baseline in the detection-power study. Wang and Isola [7] decompose contrastive embedding quality into *alignment* (paired texts land together) and *uniformity* (everything spreads out); **ambit** is unsupervised, so it measures only the uniformity half; §11 shows the same split in the measured results.

Embedding-quality scores. RankMe [9], α -ReQ [10], and their relatives (surveyed in [11]) score an embedding, without labels, by how evenly its variance spreads across directions. On unit vectors these all read the same underlying object, the covariance spectrum, so they mostly restate one another; the Vendi score [12, 13] escapes the family by counting effective modes

through a nonlinear kernel [14, 15]. **ambit** takes the lesson as a design filter (§6.5): rather than add another reading of the spectrum, it measures different objects — the pair-distance distribution and per-item fields.

Projections and density. Two-dimensional layouts preserve neighborhoods but not density [43, 44]: a visually dense region of a projection need not be dense in the embedding, which is why **ambit** draws projections but never measures on them. Kernel density estimation converges at $O(n^{-4/(4+d)})$, a rate that becomes meaningless when d is in the hundreds [31]; the per-item layer therefore rests on the distance to a measure instead. Local intrinsic dimension estimators [41, 42] are used only per item, and only to compare items with one another.

4 Experimental Methodology

Every measured claim in this report follows the protocol stated here.

Datasets. Claims are tested against synthetic data, where the answer is known by construction, and against public corpora, which are representative of real-world data. The public side splits into two roles — one corpus studied in depth, and five retrieval collections whose human relevance judgments score **ambit**’s predictions judgments — so three bodies of data appear below.

(i) *The synthetic benchmark family — known ground truth by construction.* Detection, localization, and recovery claims can only be scored against a known defect: one whose membership and scales are known exactly. The way to obtain one is to build the corpus and insert the defect deliberately, choosing which items belong to it and how tightly they sit — a *planted* pathology. Ground truth then comes from the construction, and the instrument still never consumes a label. One instance of this family is followed end to end through the report; we call it the *traced* benchmark. It holds $n = 4,000$ points at $d = 768$ (a contemporary encoder width), drawn as normalized standard Gaussians, which places them exactly uniformly on $\mathbb{S}^{d-1}$. The planted group — a *pocket* — replaces the last $k = 200$ points (5%) with $x_i = (c + s g_i) / \|c + s g_i\|$ for one shared uniform direction c , i.i.d. Gaussian g_i , and spread $s = 0.01$. That spread makes the pocket a group of near-duplicates: the median cosine inside it is 0.929, so a retrieval system cannot rank its members apart. Yet every global average stays essentially unmoved (§2.3). A defect this severe and this invisible is a stringent test for a local diagnostic. The uniform control is the same draw without the replacement. The traced benchmark is one fixed instance. A wider family of variants changes one property at a time: pocket count (1–20), size (20–800), and tightness ($s = 0.005$ – 0.06 , spanning near-duplicate to loose association). It adds mean-shift cones, low-rank collapses, and pure nulls. Together these form the test set behind the sensitivity, calibration, and power studies. All generators are stated in full above or beside their experiments; nothing is tuned to the instrument.

(ii) *A public real corpus — realistic geometry at scale.* Synthetic corpora cannot show the learned-embedding pathologies that motivate the instrument. The case study (§9) therefore uses *Nemotron-Pretraining-Legal-v1* [61], a public (CC-BY-4.0) legal corpus published by NVIDIA. From four of its thirteen subsets — the California Code of Regulations, case-law

Table 1: Evaluation corpora. All are public; the first four are used in their BEIR form [51].

corpus	documents	judged queries	task
TREC-COVID [52]	171,332	50 (dense, graded)	biomedical search
Quora [48]	522,931	10,000	duplicate matching
Natural Questions [54]	2,681,468	3,452	open-domain QA
HotpotQA [53]	5,233,329	7,405	multi-hop QA
ESCI / Shopping Queries [55]	1,215,854	8,956 (graded)	product search
Nemotron-Legal [61]	1,000,000	1,000 + 1,496	statute/case retrieval

Table 2: Encoders. Dimensions are output embedding widths; all corpora are embedded with each model’s documented document/query prompt conventions.

encoder	params	dim	lineage
all-MiniLM-L6-v2 [56, 49]	22M	384	distilled BERT
all-mpnet-base-v2 [56, 50]	110M	768	MPNet
E5-base-v2 [57]	110M	768	weak-supervision contrastive
Arctic-Embed-m-v1.5 [59]	110M	768	retrieval-specialized
EmbeddingGemma-300m [60]	300M	768	Gemma
BGE-large-en-v1.5 [58]	335M	1024	C-Pack
Qwen3-Embedding-0.6B [47]	0.6B	1024	instruction-tuned decoder

summaries, CaseHOLD, and eCFR question–answer pairs — we take 10^6 documents. We embed them at $d = 1024$ with the Qwen3-Embedding-0.6B instruction-tuned encoder [47], L2-normalize them, and store them as sharded Parquet with stable document ids. It supplies what the benchmark cannot: a real anisotropy cone and *diffuse* within-subset crowding rather than an inserted pocket.

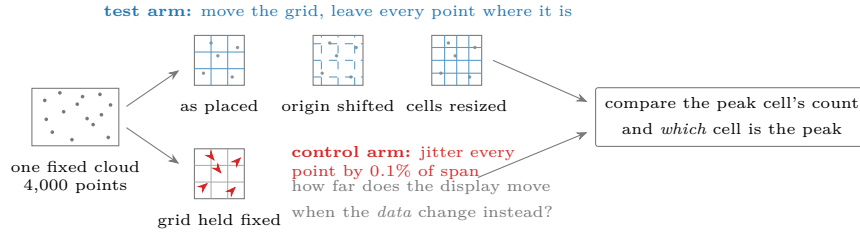
(iii) *Five public retrieval collections with relevance judgments.* The multi-corpus results (§11) and the retrieval-facing experiments (§10.5–§10.6) use five public retrieval collections with relevance judgments and seven public encoders, listed with citations in Tables 1 and 2. Judgments are quarantined: every readout is computed and hash-frozen before any judgment file is opened, and judgments are consumed exclusively by scoring code. The three do different jobs: the benchmark makes claims falsifiable, the corpus makes them representative, and the public collections score them against human judgments.

Standard settings. Every statistic runs with a fixed default configuration unless an experiment sweeps it. The defaults are stated beside each statistic where it is defined (§6–§7), and each is validated by a sweep reported in the section that uses it; Appendix Appendix A collects them in one table.

Per-experiment protocols. Each experiment below opens with one sentence saying what it exists to establish, then states the hypothesis under test, the design and the reason it takes that shape, the control it is read against, and the measure that decides. A schematic of the design accompanies each; the schematics carry no data. Full parameter values appear beside each result.

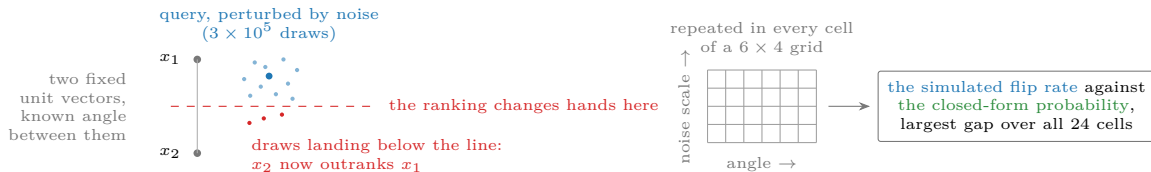
Lattice fragility (Table 3). Establishes why *ambit* refuses binned displays: the numbers they report belong to the bin placement.

- **Hypothesis:** the headline statistics of binned displays are properties of the lattice rather than of the data.
- **Design:** hold 4,000 synthetic 2-D points fixed and perturb only the lattice, shifting its origin by fractions of one cell and changing its column count. Any change in the display can then only come from the lattice.
- **Control:** an isotropic jitter of the *data* by 0.1% of their span, which shows how far the display moves when the data themselves change by a comparable amount.
- **Measure:** the change in the display's peak cell count and peak location under each perturbation, lattice against control.



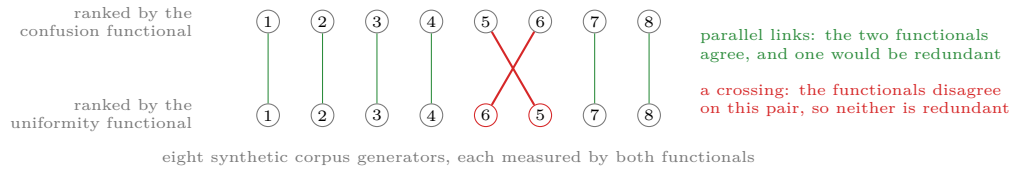
Confusion-law verification (Table 5). Establishes that the formula converting distances to collision probabilities is exact, so everything built on it inherits no approximation error.

- **Hypothesis:** the closed-form probability that noise flips a ranking is exact in any dimension.
- **Design:** two fixed unit vectors at a known angle, with 3×10^5 independent noise draws per cell of a 6×4 grid of angles and noise scales. This simulates the law's own setting directly, so there is no modeling assumption an error could hide behind.
- **Control:** the closed form itself; under the hypothesis the residual is pure Monte-Carlo noise of predictable size.
- **Measure:** the maximum absolute deviation between empirical flip rate and formula across all 24 conditions.



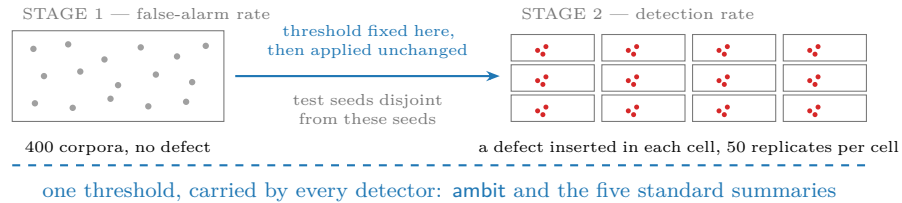
Redundancy of the confusion and uniformity statistics (Table 6). Establishes that ambit’s collision statistic is not just a restatement of the existing uniformity loss.

- **Hypothesis:** the two statistics may order corpora identically, making one redundant.
- **Design:** eight synthetic corpus generators spanning the pathology families, both statistics computed on each — redundancy is a claim about *orderings across corpora*, so the design produces orderings rather than values.
- **Control:** perfect rank agreement, the outcome under full redundancy.
- **Measure:** the rank correlation, and each individual disagreement, which is the more informative of the two.



Calibration and detection power (§8). Establishes the two properties any detector needs: it stays quiet on healthy data, and it finds defects that standard statistics miss.

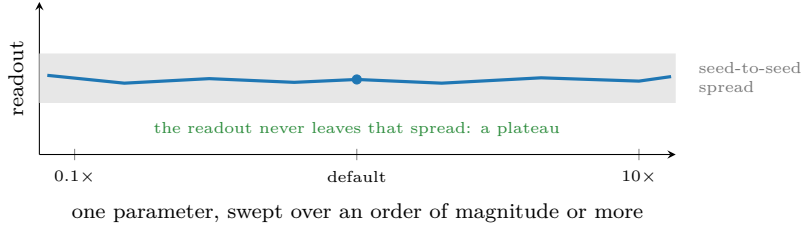
- **Hypothesis:** the instrument’s alarms fire at their nominal rates on healthy corpora and detect planted pathologies that standard summaries miss.
- **Design:** first, 400 corpora with no defect (*pure nulls*), used to measure false-alarm rates and to set every detector’s threshold at the same 1% specificity, since a power comparison is fair only when every detector is equally reluctant to fire; then a grid of planted pathologies, 50 replicates per cell, with threshold seeds disjoint from test seeds.
- **Control:** the pure nulls, and the standard summaries (mean cosine, isotropy scores, effective rank, hubness, a neighbor-distance quantile) run under identical thresholds.
- **Measure:** empirical false-alarm rate against nominal; detection rate per pathology cell at matched specificity.



Sensitivity sweeps. Establishes that no conclusion in this report depends on a lucky choice of parameter.

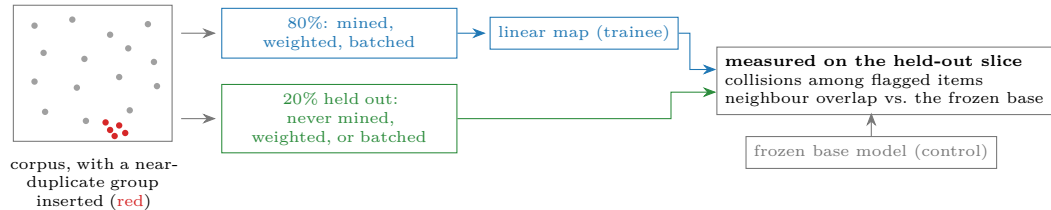
- **Hypothesis:** every default parameter sits on a plateau, so reasonable variation does not change conclusions.

- **Design:** sweep each parameter (reservoir size, pair-sample size, mass fraction, minimum pocket size, null-replicate count) one at a time over an order of magnitude or more.
- **Control:** the statistic that the swept parameter feeds — σ^* for the sample sizes, the per-document field for the mass fraction, the discrepancy z for the replicate count — computed at the default value, with seed-to-seed spread as the scale of comparison.
- **Measure:** readout drift across the sweep, compared to that spread.



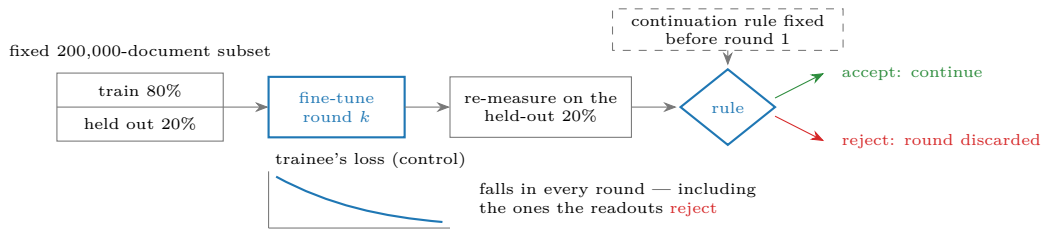
The measurements as the training objective (Table 10). This is the first of two ways the measurements can enter training. Here they are the *objective itself*: the thing being minimized, with no other loss present. In the guardrail experiment below they are instead added to a supervised training objective. Establishes what the measurements can and cannot do as a loss.

- **Hypothesis:** penalties built from the measurements reduce retrieval collisions without destroying the embedding’s structure.
- **Design:** a planted pocket with a linear map as the smallest possible trainee, and a 20% slice held out from mining, weighting, and batching — verdicts must come from data the training machinery never touched.
- **Control:** the corpus before training, and a structure-preservation score against the frozen base.
- **Measure:** expected collisions among the *flagged* items — the documents whose base-model collision counts mark them as most crowded — measured on the held-out slice, and neighbor overlap, per schedule.



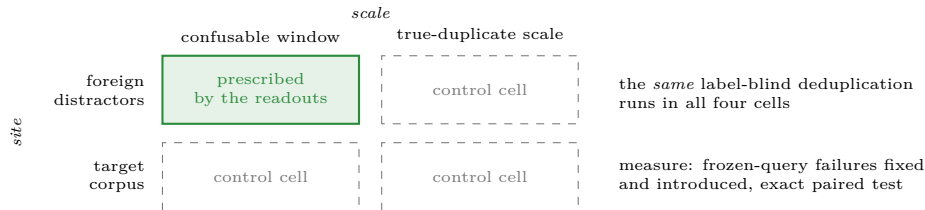
The encoder loop (§10.3). Establishes whether held-out measurements alone can accept or reject fine-tuning rounds that labels would otherwise judge.

- **Hypothesis:** held-out geometric readouts alone, with no labels, can judge fine-tuning rounds: accept the safe ones, reject the harmful ones.
- **Design:** a fixed stratified 200,000-document subset with 20% held out, and a continuation rule registered before any round ran, so acceptance cannot be negotiated after seeing results.
- **Control:** the base model’s readouts, and the trainee’s own loss curve — the signal a practitioner would otherwise trust.
- **Measure:** per-round held-out σ^* , flagged-cohort collisions, and neighbor overlap against the rule.



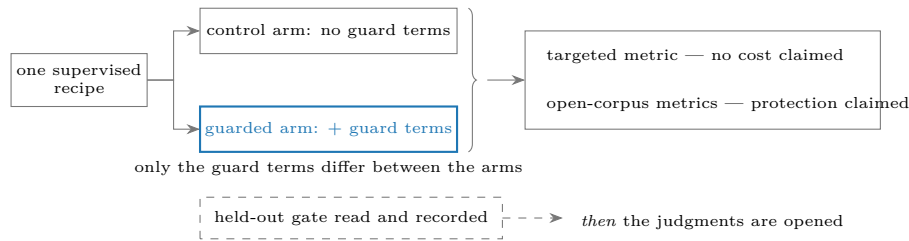
Routing (§10.5). Establishes that the measurements pick the right repair at the right similarity, and that the same repair elsewhere does harm.

- **Hypothesis:** a repair succeeds only at the site and scale the measurements prescribe — not because deduplication is generically good.
- **Design:** run the same label-blind deduplication in all four combinations of two sites (foreign distractors; the target corpus) and two scales (the confusable window; the duplicate scale). Only the full 2×2 separates “the prescription was right” from “any cleanup helps.”
- **Control:** the other three combinations, which the measurements do not prescribe.
- **Measure:** frozen-query failures fixed and introduced per cell, with an exact paired test.



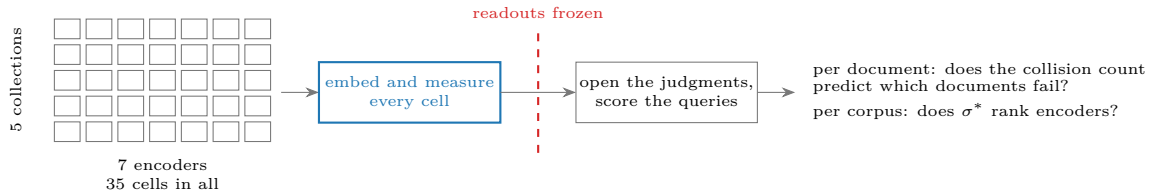
The measurements as a supplement to a supervised objective (§10.6). The second of the two ways the measurements can enter training: added to a supervised recipe as extra terms, rather than replacing its objective.

- **Hypothesis:** extra penalty terms built from the measurements — *guard terms* — prevent the damage supervised fine-tuning does outside its training distribution, without reducing its gains.
- **Design:** one supervised recipe run twice, the arms differing *only* in the guard terms, so any outcome difference is attributable to them; the held-out measurement gate is read and recorded before any judgment.
- **Control:** the unguarded arm.
- **Measure:** the retrieval metric the fine-tune targets, where the claim is no cost; the open-corpus metrics it does not target, where the claim is protection; and the gate’s pre-label prediction.



The multi-corpus, multi-encoder runs (§11). Establishes whether the results above survive outside the corpus and encoder they were developed on. These are the same measurements run over every pairing of five collections with seven encoders — thirty-five runs of one protocol rather than a new experiment.

- **Hypotheses:** first, that each document’s expected-collision count predicts which judged-relevant documents fail at their own queries; second, that the corpus-level σ^* ranks encoders within a corpus.
- **Design:** every corpus–encoder cell is embedded and fully measured, and the readouts hash-frozen, *before* any judgment file is opened. The freeze makes leakage from labels into measurements mechanically impossible rather than merely forbidden by policy.
- **Control:** for the per-document claim, the natural geometric baseline (nearest-neighbor cosine); for the corpus-level claim, mean pair cosine.
- **Measure:** per-cell prediction quality (AUC) for the first; per-corpus rank correlation with nDCG@10 for the second.



Reporting conventions. Three conventions hold throughout. Wherever a simulated no-crowding corpus exists, its value is printed next to the measured one, so every number arrives with its point of comparison. Negative results get the same prominence as positive ones. And headline numbers are replicated — benchmark results over 20 random seeds, reported as mean $\pm$ standard deviation; calibration and power studies at the replicate counts stated with each. A number shown without a spread is deterministic. Figures fall into three kinds, labelled here so a reader always knows which one is in front of them.

- **Real-corpus figures:** the crowding curve, the DTM field, and the collision curves plot Nemotron-Legal measurements, and two figures reproduce the instrument’s own report readouts on that corpus natively (the named-document localization list and the pockets bar, with the report’s values verbatim).
- **Self-computed demonstrations:** the lattice demonstration’s highlighted peak cells are computed from the figure’s own points.
- **Illustrative examples:** the remaining figures carry no data and are captioned as illustrative.

5 The Fragility of Binned Views

Binned density views — hexbin lattices over a 2-D projection, voxel grids, radial shells, fixed- k neighbor counts — are the standard tools for localizing density. **ambit** uses none of them, and this section measures why. Each of these views works by dividing the space into cells and counting how many items land in each. The trouble is that the cells are a choice — where their boundaries fall, and how large they are — and that choice changes the counts. Two analysts with the same corpus and different grids can report different crowded regions, and the gap between their answers is larger than the defects worth detecting.

5.1 The grid changes the answer

Figure 5 and Table 3 measure both choices on the hexbin display that **ambit** itself renders [26]: where the cell boundaries fall, and how large the cells are. Moving the lattice half a cell moves the headline peak count from 310 to 403, and moves which cell holds the peak (the *zonation* effect). Changing the cell size varies the peak count by a factor of 2.4 across reasonable column counts (the *scale* effect). Neither perturbation touches a single data point. In the classical extreme of the same phenomenon, re-drawing the zones over fixed data can move a correlation almost anywhere in $[-1, +1]$ [25].

Binning also wastes data: a histogram converges more slowly with sample size than a smooth estimator does [28, 29], so the partition costs precision as well as reliability. **ambit** retains one hexbin *display*, averaged over many lattice placements to remove the placement dependence [30], and captioned as a view rather than a measurement.

5.2 Every bin commits to one scale

Crowding appears at different similarity levels. Near-duplicate pockets sit above $\cos \approx 0.9$, topical clusters nearer $\cos \approx 0.5$, and the anisotropy cone spreads across the whole range.

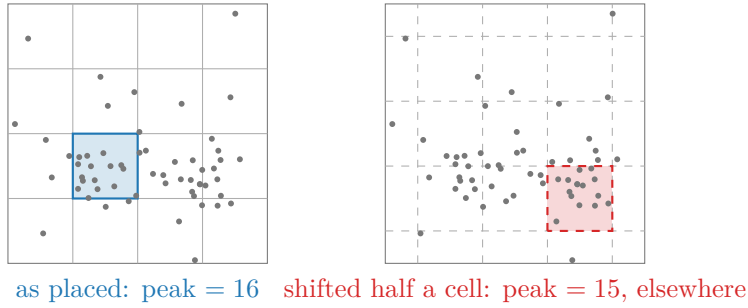


Figure 5: The modifiable areal unit problem on fixed data (the diagram’s points and high-lighted cells are computed from the data rather than drawn by hand): shifting the same lattice by half a cell changes the peak count *and relocates which cell is the peak*. Table 3 gives the effect at scale on the tool’s own hexbin display.

Table 3: Lattice fragility on fixed data ($n=4,000$ synthetic 2-D points; the tool’s own hexbin assignment). The first two blocks perturb only the instrument — its placement, then its cell size — while the last row perturbs the data instead. The text reads the comparison.

perturbation	peak-cell count	occupied cells
lattice as placed	310	256
lattice shifted 0.25 cell	310	253
lattice shifted 0.5 cell	403	262
20 columns	574	168
26 columns	340	256
34 columns	242	350
data jittered by 0.1% of span	339	—

Every bin width, shell count, and fixed neighbor count k silently commits to one of them. Inject 200 near-identical points — the benchmark’s planted-pocket size, 5% of its $n = 4,000$ — inside an already-dense cell. The hexbin peak moves from 340 to 540 — a rise that looks no different from an ordinarily busy cell. On the same corpus the continuous per-item field of §6.3 responds far more sharply: its first percentile falls by a factor of 18. A fixed k in a kNN graph is the same defect in rank-based form: in a dense pocket $k=10$ spans a tiny radius, while in a sparse outer region it spans a huge one. “Neighborhood” therefore means a different similarity level at every point, which is the drift a crowding measure must exclude.

5.3 What a cell count cannot do

Four further costs are structural.

- *Instability.* A bin count is a step function of the data: nothing bounds how far a small perturbation can move it. The field of §6.3 carries the opposite guarantee: move the corpus a little and every score provably moves only a little [32]. It holds up in practice, with self-correlation 0.997 under jitter at $d=768$.

- *No null.* A peak count carries no statement of what a well-spread corpus would show; a perfectly uniform cloud still has a hottest cell.
- *No names* (the ecological fallacy [27]). A cell count cannot say which items are crowded.
- *Projection.* Binning a 2-D layout measures the projection, not the object: far-apart groups collide in a cell, orthogonal crowding vanishes, and nonlinear layouts actively destroy density [43].

The repair is not a better density estimate: kernel density estimation needs far more points than any corpus can supply once d reaches the hundreds [31]. The repair is to drop partitions altogether and work directly with the distances between points, using functions that commit to no scale of their own and whose guarantees do not weaken as the dimension grows.

6 The Four Crowding Readouts

The measurement layer is four readouts, which together meet the four requirements of §2.3: the crowding curve, the occupancy discrepancy, the per-document crowding radius, and the catalogue of tight groups. Let $x_1, \dots, x_n \in \mathbb{S}^{d-1}$ (unit-normalized; cosine $c_{ij} = \langle x_i, x_j \rangle$; chord $r_{ij} = \|x_i - x_j\| = \sqrt{2 - 2c_{ij}}$). **ambit** reads the corpus in one pass and keeps three things:

1. exact second moments, updated per batch (these need no sampling);
2. a *reservoir* — a fixed-size uniform random subsample, maintained by replacing a random held row with decreasing probability as rows stream by — for the per-document statistics;
3. a random-pair cosine sample for the pair statistics, with a default of 2×10^5 pairs, sized so the scale readouts of this section are stable to a few thousandths.

The running figures in this section show one corpus (Nemotron-Legal, §4) so the examples stay comparable; §11 repeats the measurements on four more public collections. The traced benchmark of §2.3, whose planted pathology is known exactly, gives every layer a companion test in which the right answer is known.

6.1 The crowding curve

Definition 1 (Pair-closeness curve). $K(t) = \Pr_{(i,j)}[c_{ij} \geq t]$, hereafter the *crowding curve*: the fraction of distinct pairs at least t -similar, estimated exactly from the pair sample by sorting — no bins, no origin, no width.

$K(t)$ is Ripley’s pair-counting test (§3) carried to the sphere [16] and written in cosine coordinates. At each similarity level it reports the fraction of pairs at least that close, read against what the same number of points scattered at random would give. Two references at matched sample size accompany it (Figure 6):

(i) *The uniform null, tested globally.* For i.i.d. uniform points the pair cosine satisfies $c = 2B - 1$, $B \sim \text{Beta}(\frac{d-1}{2}, \frac{d-1}{2})$, so null pair samples are drawn directly, without constructing a single d -dimensional vector. The sphere has no boundary, so the edge corrections of §3 neither apply here nor can be misapplied [18].

Inference over the whole curve uses a one-sided global *rank-envelope* test [21]. The test works by simulation: the data curve is ranked among 199 curves drawn from the null, so the smallest attainable p is $1/200$, and a single global p covers all scales at once. Many curves tie, because every null curve is zero over most of the range. Ties are broken by how extreme each curve becomes at its most extreme point (its *extreme rank length*). The α -critical envelope is the band the null curves span at the test’s significance level, and *liftoff* is the highest cosine at which the data escape it. Liftoff is annotated only when the global test rejects. On 40 pure-null corpora the gated test reported 0 liftoffs. The naive pointwise band (the max of the null curves) fired on *every* null seed at the bulk edge — a direct demonstration of the multiple-comparison failure — and even the α -critical envelope alone over-fires $\approx 4\times$ nominal. The division of labor is therefore strict: the envelope locates the scale, and the rank p decides significance.

(ii) *The anisotropy-conditioned null.* Real corpora are cone-shaped for benign reasons, so judging them against an even spread alone reports a defect where there is none. The second reference is the *angular central Gaussian* (ACG) of directional statistics [20], built from the corpus’s own covariance spectrum: $x = \Sigma^{1/2}g/\|\Sigma^{1/2}g\|$, which rotation invariance lets us sample from the eigenvalues alone. It reproduces the corpus’s cone without its clustering, so any height of the data curve above it is crowding that the cone cannot explain.

Figure 6 shows the curve on the real corpus (Nemotron-Legal, §4): the global test rejects at the minimum attainable $p = 1/200$ with liftoff at $\cos \approx 0.81$. Through mid-similarities the data track the anisotropy-conditioned reference, so that part of the excess over uniformity is the cone itself, benign and expected of a learned encoder. Above $\cos \approx 0.8$ the data rise above both references: genuine tight-pair structure at the near-duplicate scale, which §6.3 attributes to named documents. On the traced benchmark the same test rejects at $p = 1/200$ with liftoff $\cos \approx 0.93$ and a shoulder of excess mass exactly matching the planted pocket’s pair share (2.4×10^{-3}) — the counterpart with known ground truth. The lower panel of Figure 6 repeats the reading across six corpora under one encoder. Every corpus carries tight-pair mass far above the null, but the anatomy differs: ESCI and TREC-COVID hold the most near-duplicate mass, Quora the least beyond its paraphrase band. A single scalar would average that structure away.

6.2 The occupancy discrepancy (Stolarsky z)

A binned count inspects one cell of one lattice. The complete version of that question inspects every spherical cap — every center, every radius — and asks how far each cap’s occupancy departs from the uniform expectation. A classical identity collapses this total departure into a single pairwise average.

Proposition 1 (Stolarsky [22]). *For points on $\mathbb{S}^{d-1}$, the spherical-cap L_2 discrepancy — the squared deviation between empirical and expected cap occupancy, integrated over all cap centers and all cap radii — equals, up to explicit constants, a constant minus the mean pairwise chord $\frac{1}{n^2} \sum_{i,j} \|x_i - x_j\|$.*

The mean pairwise chord therefore performs the audit no single lattice can: it covers every patch of the sphere at every size, and there is no lattice left to choose. *ambit* reports it as a z -score — the number of standard deviations by which the corpus differs from simulated no-crowding corpora of the same pair-sample size — so that sampling noise is accounted for.

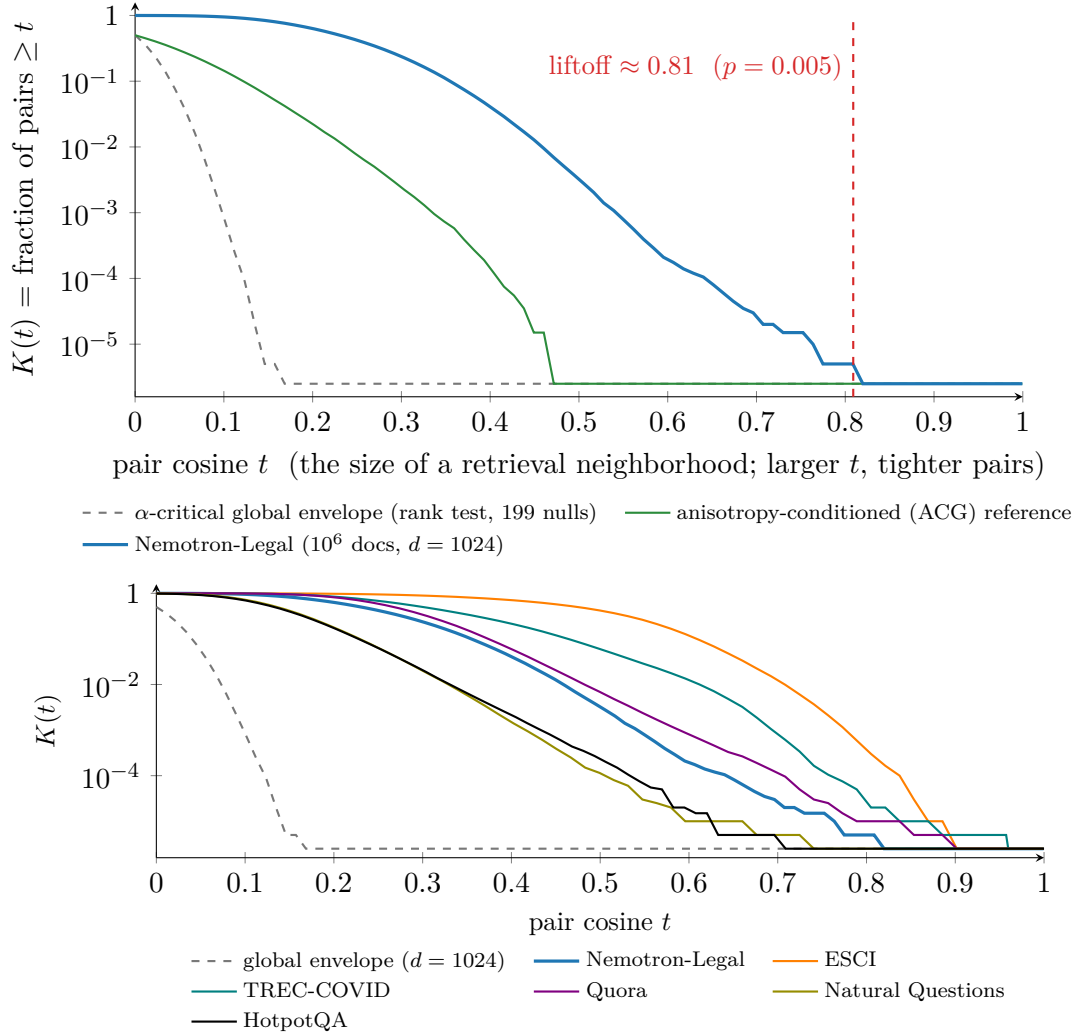


Figure 6: The crowding curve on Nemotron-Legal [61]. Top: the data track the anisotropy-matched reference through mid-similarities, then rise above both references past $\cos \approx 0.8$, where the global test rejects at $p = 1/200$. Bottom: the same curve for six corpora under one encoder, each with its own shape.

Measured on the benchmark over 20 seeds, the uniform control scores $z = -0.14 \pm 1.00$: a mean at zero and unit spread, which is what calibration looks like. Planting the 5% pocket moves the mean chord from 1.41405 to 1.41160, a shift of 0.17%, which the concentrated null resolves as $z = -47 \pm 4$. Tightening the pocket further (members within a cap of angular radius ≈ 0.15) drives the same statistic to $z = -430$: detection strength scales with the pathology while the control stays at noise level.

Two properties govern how z is read.

- **z is a test statistic rather than an effect size.** Its magnitude grows with the pair-sample size P , because the null mean’s spread shrinks as $1/\sqrt{P}$. Values are comparable only at matched P , and the instrument reports them at its standard $P = 2 \times 10^5$. The

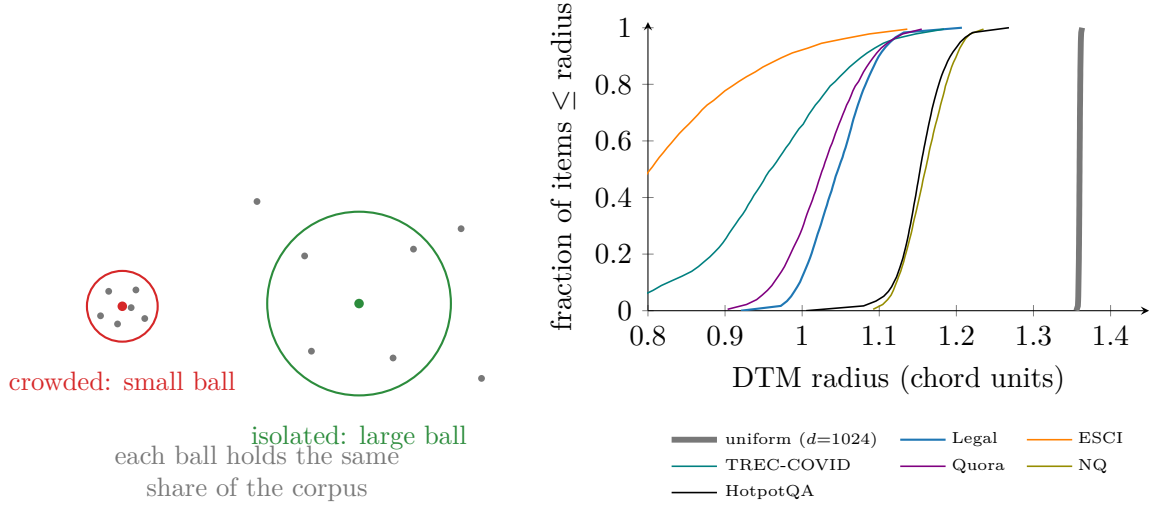


Figure 7: The distance-to-measure field. *Left*: an illustrative example — each item is scored by the radius of the ball it needs to gather a fixed share ($m = 2\%$) of the corpus. *Right*: the field’s exact CDF on the real corpus (measured): the *entire* field sits left of the uniform spike at 1.36, and the finding is the low tail against the corpus’s own bulk (1st percentile 0.966 against median 1.047), whose members the instrument names by id (Figure 8); the text explains why the uniform band reads on the cone rather than on pockets. The five grid corpora under the same encoder overlay their CDFs: each has its own bulk position and low-tail mass — per-item crowding profiles differ far more across corpora than any global statistic suggests.

scale readouts σ^* and liftoff behave like effect sizes instead, concentrating as $1/\sqrt{P}$: $\text{sd}(\sigma^*)$ falls from 0.0126 to 0.0013 as P grows from 10^4 to 10^6 .

- **The null’s own spread must be estimated well.** With 24 null replicates $|z|$ is inflated by $\approx 20\%$ and reproduces with $\text{sd} \approx 8$. The default of 64 replicates sits where that inflation has died away; there the benchmark reads $z = -43 \pm 4.5$.

6.3 Scoring how crowded each document is (the DTM field)

Definition 2 (Empirical DTM [32, 33]). *For mass fraction m and $k = \lceil mn \rceil$: $\text{DTM}_m(x_i) = \left(\frac{1}{k} \sum_{j=1}^k r_{i,(j)}^2 \right)^{1/2}$, with $r_{i,(j)}$ the distance to x_i ’s j th nearest reservoir point — the radius the item needs to gather an m -share of the corpus.*

A small radius means the item is crowded; a large radius means it is isolated (Figure 7). The single parameter m is a *mass fraction*: a share of the corpus rather than a count, so it needs no adjustment as n grows. It replaces bin width, grid origin, and fixed k at once. Its one trade-off is that m sets the smallest group the field can resolve. Table 4 measures the boundary: separation between a planted pocket and the bulk collapses once m exceeds the pocket’s share of the corpus, k/n . The reporting rule follows — choose m below the smallest pocket share worth catching — and the default $m = 0.02$ resolves pockets down to 2% of the corpus.

Table 4: The DTM mass fraction’s measured blur boundary (bulk median / pocket median of the field; benchmark family, planted pocket of k members in $n = 4,000$). Separation survives while $m \lesssim k/n$ (boldface) and collapses beyond it — exactly the predicted rule.

	$m=0.005$	$m=0.01$	$m=0.02$	$m=0.05$	$m=0.1$
$k = 20$ ($k/n=0.005$)	2.83	1.34	1.13	1.05	1.02
$k = 50$ ($k/n=0.013$)	3.63	3.59	1.52	1.14	1.06
$k = 200$ ($k/n=0.050$)	3.70	3.69	3.67	3.52	1.37
$k = 800$ ($k/n=0.200$)	3.75	3.74	3.74	3.72	3.70

most crowded documents

id · DTM radius · $\approx$ collisions at $\sigma^*=0.12$

9b3dd305-0e50-4...	0.921	≈ 12
a310931d-78cc-4...	0.922	≈ 12
369ba45c-63b5-4...	0.923	≈ 4
7ce46beb-eb01-4...	0.928	≈ 4
18a8b3f6-a37c-4...	0.933	≈ 3
efaeae26-62ef-4...	0.936	≈ 4

most isolated (voids)

59c2a211-f8cc-4...	1.207
2a761d0e-c3e0-4...	1.198
0eba585f-60f9-4...	1.190

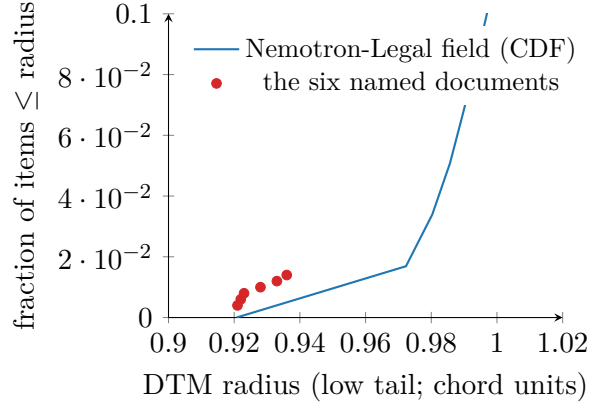


Figure 8: The per-entity localization readout, reproduced natively from the instrument’s report on Nemotron-Legal (values are the report’s own). *Left*: the crowded-document list — named by *uuid*, each with its DTM radius and its expected collision count at the corpus’s σ^* : the radius says how crowded, the count says what it costs. *Right*: where those six sit — the extreme low tail of the field’s CDF (the bulk rises to 1 near radius 1.05; the uniform band lies far right at 1.36). Aggregate, cell-based displays cannot provide this output.

The estimator also inherits a guarantee that step functions lack: perturb the corpus a little and every item’s score provably moves only a little, bounded by $\|\text{DTM}_m^P - \text{DTM}_m^Q\|_\infty \leq m^{-1/2} W_2(P, Q)$ [32]. It holds in practice too — jittering every coordinate by 10^{-3} leaves the field 0.997-correlated with itself. On a fixed 10^6 -row *synthetic* corpus (the benchmark generator scaled up, with the 5% pocket planted), the field’s first percentile does not depend on which reservoir is drawn: it spans 0.365–0.366 across reservoirs of 10^3 – 1.6×10^4 rows, ten seeds each. Every other headline readout is equally stable; σ^* spans 0.0559–0.0562 against 0.0564 at a 3.2×10^4 reference. On the traced benchmark the planted pocket drags the low tail to 0.364 against a bulk at 1.35 — the $3.7 \times$ separation behind the detection-power results of §8.

Both tails are actionable and *named*: the low tail is the crowded-item list, each entry annotated with its expected collision count at the corpus’s own σ^* (§7), and the high tail is the list of coverage voids. In strongly cone-shaped corpora the cone shortens every radius, so the entire field sits below the uniform band. When that happens the instrument says so, and the finding becomes the low tail read against the corpus’s own bulk rather than against the uniform band.

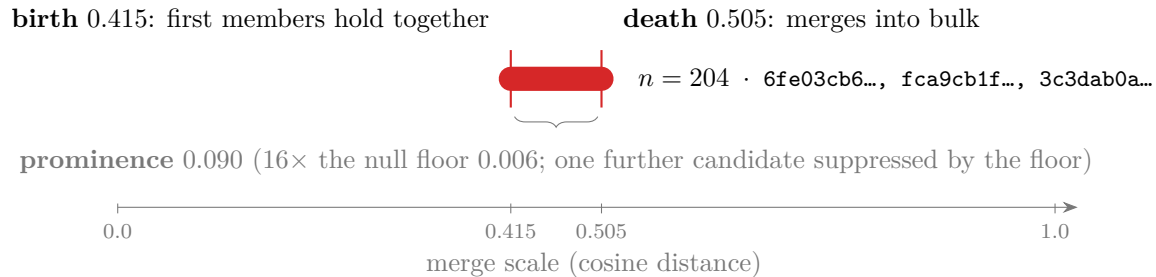


Figure 9: The pockets readout, reproduced natively from the instrument’s report on Nemotron-Legal with the merge-tree anatomy annotated on it: the one pocket surviving the null-calibrated prominence floor — the largest prominence pure noise produces at matched size and dimension — (204 members, sample ids on the bar) is *born* when its first members hold together and *dies* into the bulk (excess-of-mass selection). Its prominence 0.090 is modest — a loose association rather than a duplicate cluster — yet 16× the null floor, and therefore reportable. No flat threshold enters anywhere in the construction.

6.4 Finding the tight groups (the merge tree)

Between the global curve and the per-item field sits *structure*: whether the corpus forms one broad pile or many tight pockets, and which documents constitute each. *ambit* answers with a merge hierarchy, built by single linkage on a modified distance (the *mutual-reachability* distance [35]): each pair’s distance is replaced by the largest of three numbers — the pair distance itself, the first point’s distance to its own k th neighbor, and the second point’s — so that one stray point cannot bridge two groups. The tree is condensed at a minimum pocket size, and pockets are selected by excess of mass in $\lambda = 1/\text{distance}$: on each branch, the cluster persisting over the widest scale range is retained. In the vocabulary of persistence theory this is H_0 persistence with the *elder rule* — when two groups merge, the more recently formed one is the one considered ended [34, 36]. Each selected pocket reports birth, death, prominence (death minus birth: the scale range over which it persists as a separate entity), size, and member ids (Figure 9). Three implementation findings matter.

- **Elder-rule inversion.** The naive elder rule is wrong for this purpose: the tightest pocket is the *eldest* component and survives to the root, so the bulk “dies into it.” Excess-of-mass selection resolves the inversion.
- **Caterpillar membership.** Single-linkage chains inside a tight pocket are caterpillars (long chains with single-node side branches). Membership is therefore reported as the condensed core plus the side branches that fell out in the tight regime (below the geometric mean of birth and death), excluding bulk stragglers.
- **The childless root.** Chaining can prevent *any* real split from existing: a tight pocket may seed the one component that then absorbs the bulk point by point, so no node ever has two children of the minimum size, the only cluster is the root, and the tightest structure in the corpus goes unreported. A parameter sweep exposed this failure — at minimum size 32 a planted 200-member pocket returned nothing, and 3 of 20 benchmark seeds missed it even at the default. The fix treats a childless root as a candidate with

death at the final merge scale, whereupon the membership rule extracts exactly the pocket core.

With the fix in place, the tree recovers the planted pocket in 20 of 20 seeds at exact size (200.0 ± 0.0 ; birth 0.063, death 0.898, prominence 0.835) across min-sizes 8–64, and on a two-pocket variant recovers sizes 60 and 40 exactly.

Reported prominence carries a *null floor*. Uniform corpora of matched size and dimension produce maximum prominences of only ≈ 0.005 (99th percentile 0.009 over 400 nulls), so the instrument simulates that floor on every run and suppresses any pocket within it. A genuine pocket sits two orders of magnitude above the floor. Figure 9 shows the resulting readout on the real corpus: one pocket survives the floor (204 members, sample ids on its bar), and one further candidate was suppressed.

6.5 What the measurements cost, and what was excluded

All continuous-layer statistics run on the reservoir or the pair sample. The curve and its nulls are $O(P \log P)$ in the pair-sample size. The DTM field is a blocked Gram top- k , $O(n_r^2 d)$ on a capped subsample. The merge tree is a dense Prim MST on $n_r \leq 4,096$ rows plus a linear condensation pass. Full-corpus second moments stream in one pass.

Candidate metrics were admitted by five filters. They are recorded here because they make future proposals cheap to evaluate.

- *Spectrum*: does it reduce, under a linear kernel, to the covariance spectrum already reported?
- *Continuity*: is it a continuous function of the data, or a step function of bin, rank, or partition membership?
- *Null*: does it come with a calibrated reference?
- *Naming*: can it score individual items, or only regions?
- *Supervision*: is it label-free?

These excluded, among others: RankMe, NESum, and stable rank (spectrum); k -means-conditioned statistics (continuity); raw counts (null); all cell aggregates (naming); and neural-collapse metrics [46] (supervision).

7 Operational Semantics: the Resolution Bandwidth

The measurements so far say how evenly a corpus covers the sphere. A practitioner needs something else: how imprecise a query may be before the wrong document wins. An exact formula connects the two, and simulation confirms it.

7.1 An exact pairwise confusion law

A query that intends document x lands near it rather than on it. The simplest model of that is $q = x + \sigma g$ with $g \sim \mathcal{N}(0, I_d)$, where σ sets how far the query strays. A competitor y then wins — a *collision* — whenever $\langle q, y \rangle > \langle q, x \rangle$.

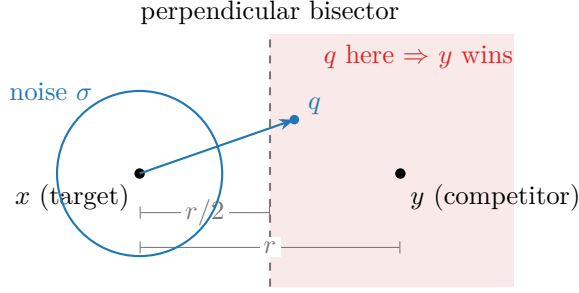


Figure 10: The confusion law’s geometry: the decision boundary between target and competitor is the perpendicular bisector at distance $r/2$; Gaussian noise of scale σ crosses it with probability exactly $\Phi(-r/2\sigma)$. A far competitor is safe; a near-duplicate ($r \rightarrow 0$) has error probability approaching $1/2$ at every noise scale.

Table 5: Verification of Proposition 2 by direct simulation at $d = 768$. Each cell uses 3×10^5 trials, enough to resolve deviations at the few- 10^{-4} level of the Monte-Carlo error. Empirical flip rate vs. $\Phi(-r/2\sigma)$ with $r = 2 \sin(\theta/2)$. Maximum absolute deviation across all 24 conditions: 9×10^{-4} , i.e. Monte-Carlo error — the law is exact rather than asymptotic.

	$\sigma = 0.05$		$\sigma = 0.15$		$\sigma = 0.30$		$\sigma = 0.60$	
θ	emp.	pred.	emp.	pred.	emp.	pred.	emp.	pred.
0.1	.1583	.1588	.3700	.3695	.4335	.4338	.4677	.4668
0.3	.0014	.0014	.1592	.1596	.3101	.3092	.4021	.4017
0.6	.0000	.0000	.0245	.0244	.1626	.1623	.3113	.3112
1.0	.0000	.0000	.0007	.0007	.0554	.0550	.2114	.2121
1.4	.0000	.0000	.0000	.0000	.0156	.0159	.1416	.1415
2.0	.0000	.0000	.0000	.0000	.0024	.0025	.0803	.0804

Proposition 2 (Exact flip law). *For unit vectors at chord distance $r = \|x - y\|$, $\Pr[\langle q, y \rangle > \langle q, x \rangle] = \Phi(-r/2\sigma)$, exactly, in any dimension.*

Proof. Geometrically (Figure 10): the set of queries for which y beats x is the half-space beyond the perpendicular bisector of x and y . The query starts at x , at distance $r/2$ from that boundary, and isotropic Gaussian noise carries it across with probability $\Phi(-(r/2)/\sigma)$. Algebraically: $\langle q, y - x \rangle$ is Gaussian with mean $\langle x, y - x \rangle = \cos \theta - 1 = -r^2/2$ and standard deviation $\sigma \|y - x\| = \sigma r$, so the flip probability is $\Phi((-r^2/2)/(\sigma r)) = \Phi(-r/2\sigma)$. $\square$

Simulation confirms the law across a grid of angles and noise scales (Table 5). One consequence matters most. When two documents are nearly identical ($r \rightarrow 0$), a query aimed at either returns the wrong one about half the time, at every noise scale, however small. No amount of query precision rescues them. This is why duplicate pockets are the worst crowding pathology, and why the crowding curve’s liftoff at the near-duplicate scale (§6.1) matters operationally.

7.2 $C(\sigma)$, σ^* , and the uniformity bound

Summing the law over competitors gives the expected number of items outranking the intended target,

$$C(\sigma) = \sum_{j \neq i} \Phi\left(-\frac{r_{ij}}{2\sigma}\right) \approx (n-1) \int \Phi\left(-\frac{r}{2\sigma}\right) dK(r),$$

a quantity that depends only on the distances between pairs of documents. Adding up the individual probabilities can only overstate the chance that at least one competitor wins, no matter how those events correlate (the union bound). C therefore errs on the high side. Hubness — a few items recurring as near neighbors of many others — correlates the events, which can push the bound further above the truth but never below it.

Definition 3 (Resolution bandwidth). σ^* is the largest σ with $C(\sigma) \leq 1$ — the corpus’s tolerance for imprecise queries, the most noise a query can carry before one wrong item is expected to outrank the right one. C is monotone, so bisection inverts it.

Figure 11 compares the real corpus with its uniform control. Nemotron-Legal crosses the one-collision tolerance at $\sigma^* = 0.123$, against 0.148 for a well-spread corpus of the same size and dimension: it retains 83% of the ideal noise budget. On the benchmark, the planted 5% pocket spends 26% of the budget ($\sigma^*: 0.203 \rightarrow 0.150$).

The scale σ is noise per coordinate, but it converts into familiar cosine units. A noisy query’s expected similarity to its own target decays as $\langle \bar{q}, x \rangle \approx (1 + \sigma^2 d)^{-1/2}$. At $d = 1024$, $\sigma^* = 0.123$ therefore means a query may drift until its cosine with its target falls to ≈ 0.25 and still be expected to win.

The lower panel overlays six corpora under one encoder. Their tolerance crossings span $\sigma^* = 0.090$ to 0.129: one number per corpus for how much query noise it tolerates.

Proposition 3 (Uniformity as a Chernoff bound). With $U_t = \log \mathbb{E} e^{-t\|x-y\|^2}$ the uniformity statistic of [7]: $C(\sigma) \leq (n-1) e^{U_t}$ at $t = 1/(8\sigma^2)$.

Proof. $\Phi(-z) \leq e^{-z^2/2}$ with $z = r/2\sigma$. □

This gives a practical meaning to a setting practitioners currently choose by convention. Anyone training with the uniformity loss must pick its temperature t ; the bound shows that each choice of t corresponds to a level of query imprecision, so training at $t = 2$ is minimizing an upper bound on expected collisions at $\sigma = 0.25$. `ambit` reports uniformity with this reading attached.

7.3 Scope

The guarantee has three boundaries.

- *Query population.* The corpus is treated as its own query population. This is exact for deduplication, clustering, and corpus-as-queries retrieval; extrapolating to an external workload assumes the workload lands where the documents are.
- *Noise shape.* The model’s noise is isotropic — the same in every direction — while real query drift has structure. To test how much this matters, we re-ranked a grid of corpus

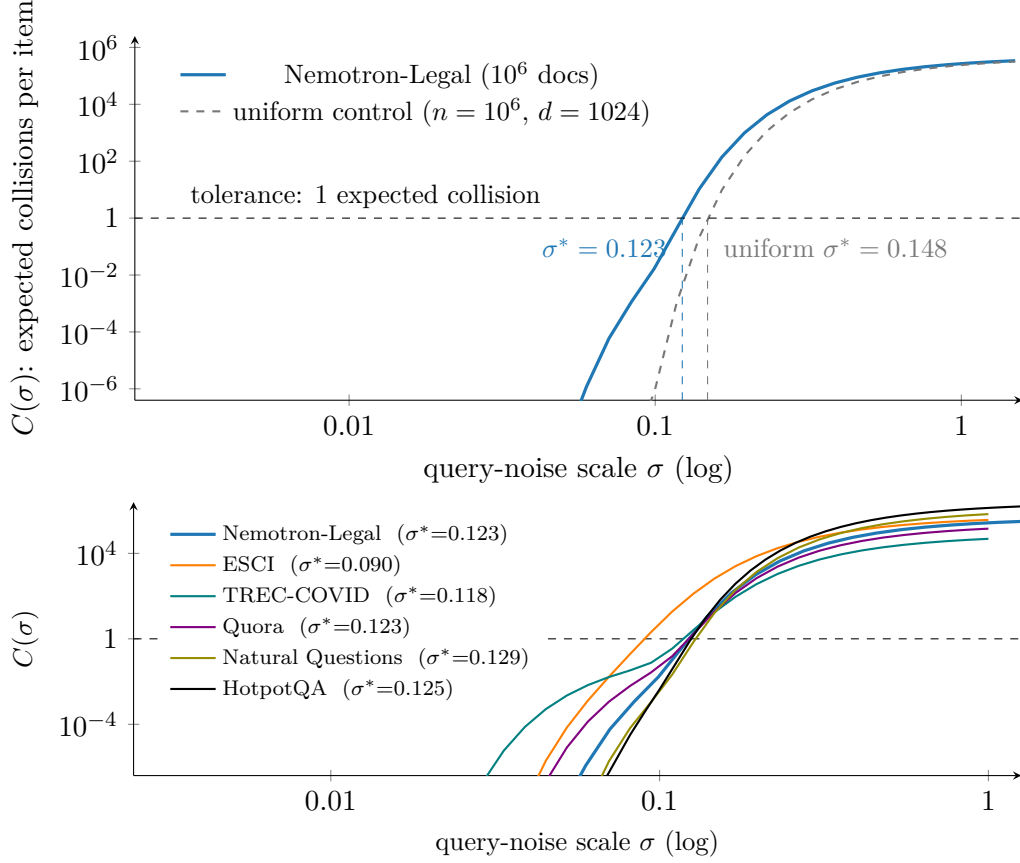


Figure 11: Expected collisions vs. query noise on the real corpus (measured; Nemotron-Legal). Top: the corpus against its uniform control — the corpus’s cone and tight pairs dominate $C(\sigma)$ at the noise scales where retrieval operates, pulling the tolerance crossing from $\sigma^* = 0.148$ to 0.123 . At large σ the curves converge (every pair confuses) and at tiny σ both vanish; the tolerance crossing is determined where they differ. Bottom: the six corpora under the same encoder, with tolerance crossings spanning $\sigma^* = 0.090$ to 0.129 ; the text gives the reading.

types under structured noise of the same overall size. The isotropic ranking survives moderately structured noise (rank-25 subspace noise: 0.99 pairwise-order agreement over five seeds). It degrades only when the noise model is badly wrong (rank-2: 0.79; noise aligned with each corpus’s own top principal directions: 0.82). Comparative use is therefore robust; noise built to align with the corpus is the stated failure mode.

- *Aggregation.* C is the quantity every later decision is priced in — σ^* is defined from it — so the direction of its bias matters. The aggregate is an upper bound: C over-counts confusions, never under-counts them, so acting on it errs toward caution.

σ^* is therefore a conservative guarantee about one corpus at a time, and it is strongest used comparatively — the same corpus embedded two ways, or the same embedding tracked over time.

Table 6: Measured near-redundancy of the confusion statistic with uniformity across eight synthetic corpus types ($n = 2,000$, $d = 256$; C at $\sigma = 0.25$, U at the matched $t = 2$). Rank correlation 0.978; 27 of 28 pairwise orderings agree. The one disagreement (boldface) is between qualitatively different pathologies — and the confusion ordering is the operationally correct one.

corpus type	$C(0.25)$	$U_{t=2}$
uniform	4.86	−3.97
one cluster of 50	4.90	−3.96
twenty clusters of 40	5.41	−3.90
one clump of 200	5.58	−3.87
five clumps of 100	5.80	−3.84
anisotropic cone	8.38	−3.48
low-rank collapse	15.49	−3.08
cone + duplicate pocket	17.78	−3.21

7.4 Negative results

Two negative results from the design process are recorded to prevent their re-derivation.

Near-redundancy of C with uniformity (Table 6). For telling corpora apart at the corpus level, the confusion transform adds almost nothing beyond uniformity. Its value lies elsewhere: it carries retrieval units, it is calibrated, and it produces per-item counts. The single disagreement in the table is the informative part. The confusion transform ranks a duplicate-pocket corpus as worse than a diffuse low-rank collapse, and the uniformity statistic ranks them the other way. Duplicate pairs confuse a query about half the time at every noise scale, while a spread subspace stays partly navigable, so the confusion ordering is the operationally correct one.

A vacuous stability certificate. We tried to turn the per-item stability theorem into finite-sample error bars by combining it with a split-half empirical Wasserstein estimate. At embedding dimension the resulting bars are useless, for a structural reason. Two independent halves of the same corpus sit about as far apart as the data scale itself (≈ 1.35 on $\mathbb{S}^{767}$ at $n_r = 2,000$). The certificate therefore comes out at $m^{-1/2}\widehat{W}_2 \approx 13.5$, an order of magnitude wider than the entire range of the field it is meant to bound. A dimension-free certificate would need a different metric carrying its own stability theorem, which is an open question rather than an engineering task.

8 Calibration and Detection Power

The instrument’s statistics were calibrated against 400 corpora built with no crowding, so that any alarm they raise is a false alarm by construction. Four hundred is the number needed to pin a 1% false-alarm rate to a standard error near 0.5%. Detection power was then compared against a suite of standard diagnostics. Every threshold, the instrument’s and the baselines’ alike, was set on those same nulls at a matched 1% false-alarm rate, so every detector is equally reluctant to fire.

Table 7: Detection power at matched 1% false alarm (50 replicates per cell, enough to estimate each power entry to a standard error of at most 0.07; thresholds calibrated on 400 nulls; $n = 2,000$, $d = 256$). Pocket cells are (count $\times$ size, spread s ; $s = 0.005$ is near-duplicate, 0.06 loose). Continuous-layer detectors left of the rule; baseline suite right. The hardest cells — a single pocket of 1% of the corpus — separate the layers; boldface marks the exceptions discussed in the text.

cell	rank _p	σ^*	DTM	pocket	meancos	Iso	unif.	hub	kNN _{q05}
1 $\times$ 20, $s=.005$	1.00	1.00	1.00	1.00	.04	1.00	.88	.18	.22
1 $\times$ 20, $s=.02$	1.00	1.00	1.00	1.00	.02	1.00	.70	.42	.34
1 $\times$ 20, $s=.06$	1.00	.22	1.00	1.00	.04	1.00	.02	.90	.32
1 $\times$ 50, $s=.005$	1.00	1.00	1.00	1.00	.58	1.00	1.00	1.00	.92
5 $\times$ 20, $s=.02$	1.00	1.00	1.00	1.00	.26	1.00	1.00	1.00	1.00
1 $\times$ 200, any s	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
cone (mean shift)	1.00	1.00	1.00	.12	1.00	.00	1.00	1.00	1.00
low-rank collapse	1.00	1.00	1.00	1.00	.26	1.00	1.00	.00	1.00

Null calibration. On crowding-free corpora ($n = 2,000$, $d = 256$; the benchmark generators with no defect planted), every alarm fires close to the rate it advertises.

- the whole-curve rank test rejects at $p \leq 0.01$ in 1.25% of them, against a nominal 1%;
- liftoff, which is reported only when that test rejects, fires in 6% at $\alpha = 0.05$, against a nominal 5%;
- the occupancy discrepancy exceeds $|z| > 3$ in 0.75%;
- the largest pocket prominence has median 0.005 and 99th percentile 0.009 — the reporting floor of §6.4.

Measured false-alarm rates for all calibrated thresholds land on 0.010–0.013.

Detection power. Table 7 reports detection power on every planted defect in the benchmark family, and three readings follow.

- **Full power on every pocket.** The continuous layer — the rank test, σ^* , the DTM tail ratio, and pocket prominence — detects every planted pocket, including the hardest cell: a single pocket of 20 items holding 1% of the corpus. On that cell mean cosine (0.04), hubness (0.18), and the kNN-distance quantile (0.22) sit near the false-alarm floor, because too little of the corpus participates to move any average. This is the claim of §2.3 in numbers.
- **The exceptions.** Covariance-spectrum statistics (IsoScore) also reach full power on a single coherent pocket. When 1% of the corpus shares one direction, that direction alone carries enough variance to stand out from the rest, so a spectrum summary is enough to detect one tight pocket at the corpus level. Three things it still cannot do: name the pocket’s members, detect a mean-shift cone (power 0.00, since centering removes the shift), or state what the crowding costs in retrieval terms.

- **The discriminating cells behave as predicted.** The occupancy discrepancy z is weak on small pockets, because a small pocket owns a tiny pair share. The pocket detector declines to alarm on a cone (0.12), which is correct: a cone is not a pocket. And hubness is blind to a low-rank collapse.

The layer’s value is therefore not one more powerful summary number. It detects everything the baselines detect, at the same false-alarm rate, and it adds three things they cannot supply: the names of the documents involved, the similarity level at which the trouble sits, and a count of the retrieval errors it implies.

9 ambit

9.1 Architecture

ambit is a library with a thin CLI, and one canonical in-memory type (`EmbeddingSet`) flows through its pipeline. A *streaming scan* makes one pass over the input — `.numpy`, `.npz`, `.parquet`, `.csv`, or `.jsonl`, as a single file or a sharded directory — accumulating exact second moments, norms, and a uniform reservoir. A context builder then derives projections (PCA by default), the eigenspectrum, a kNN graph, unsupervised clusters, and the pair-cosine sample. The graph can be filtered to *reciprocal* neighbors, which suppresses hubs by construction.

All measurements of §6–§7 are library functions over these objects, and a separate presentation layer (Appendix Appendix B) renders them. The core depends on `numpy` alone; heavier capabilities (tabular IO, ANN graphs, UMAP, GPU paths, on-the-fly embedding via an OpenAI-compatible endpoint, the training extra) are optional and degrade gracefully.

A comparison mode aligns two embeddings of the same items by id. It reports a local number — how many of each item’s nearest neighbors survive the change — beside global similarity statistics (centered kernel alignment, CKA [45], kernel and rigid distances). The two can disagree, and the disagreement is informative: high CKA with low neighbor overlap means the change kept the broad shape of the space while rewriting who sits next to whom.

9.2 Scaling

Table 8 gives the measured curves, and the practical consequence is that corpus size costs little beyond reading the bytes. A $10^7 \times 1024$ corpus of 41 GB streams in 18.6 s from raw `.numpy`; the Parquet path adds decode overhead, and the case study’s sharded-Parquet run takes ≈ 3.5 minutes end to end. Everything downstream of the scan works from the reservoir, so it costs the same at 10^5 rows as at 10^7 : context build ≈ 3 s, and the entire continuous layer ≈ 1.8 s.

Dimension is the variable that does cost. It enters through the scan’s d^2 scatter accumulation and the context build’s d^3 eigendecomposition, the one super-linear phase at 52.6 s for $d = 4096$. The continuous layer grows only through BLAS on the reservoir Gram, from 1.8 to 3.4 s over $d = 128$ to 4096.

The approximate mode carries one demonstrated hazard. With the planted pocket placed in the corpus’s *last* rows — the layout that label-sorted shards produce — the full scan detects it (rank test $p = 0.010$, liftoff 0.92). Approximate scans of the first 5×10^4 to 3×10^5 rows see a clean corpus ($p = 0.18$ – 0.45 , no liftoff, mean cosine ≈ 0). An approximate scan’s reservoir

Table 8: Scaling on the measurement host (numpy-only core; raw `.npy` input; per-phase wall time and peak RSS from forked children — RSS is dominated by memory-mapped input pages rather than allocator footprint). Left: corpus-size sweep at $d = 1024$; the streaming scan is linear in bytes at ≈ 2 GB/s and the reservoir-bound phases are *constant* in n . Right: dimension sweep at $n = 10^6$; the scan carries the d^2 scatter accumulation and the context build the d^3 eigendecomposition, while the continuous layer stays nearly flat.

$d = 1024$				$n = 10^6$			
n	scan	ctx	cont.	d	scan	ctx	cont.
10^5	0.2 s	2.7 s	1.8 s	256	0.6 s	0.5 s	1.8 s
10^6	2.0 s	3.0 s	1.8 s	1024	2.0 s	3.2 s	1.8 s
3×10^6	5.6 s	3.0 s	1.8 s	2048	4.6 s	10.4 s	2.0 s
10^7	18.6 s	2.6 s	1.8 s	4096	12.4 s	52.6 s	3.4 s

Table 9: Case study: Nemotron-Pretraining-Legal-v1 [61] (four subsets, 10^6 documents), $d = 1024$; full report in ~ 3.5 minutes on CPU, one streaming pass.

items $\times$ dims	1,000,000 $\times$ 1024
mean pair cosine	+0.236
IsoScore / effective rank	0.090 / 728.0/1024
occupancy discrepancy z	−4,211
whole-curve rank test	$p = 0.005$; liftoff $\cos \approx 0.82$
resolution bandwidth σ^* (uniform null)	0.123 (0.148)
noise budget retained	83%
top pocket (size; birth; prominence)	204; 0.41; 0.09
kNN purity / silhouette (4 provided subsets)	0.95 / +0.04
most-crowded items	≈ 3 –12 expected collisions each

is only as representative as the row order that fed it. The mode therefore suits exploratory iteration. It is unsafe for conclusions about data whose layout may correlate with its content.

9.3 Case study: a million-document corpus

Table 9 summarizes a full run on one million documents drawn from four subsets of the Nemotron-Pretraining-Legal-v1 corpus [61] (§4), embedded at $d = 1024$ by an instruction-tuned text encoder. The corpus is strongly cone-shaped (mean pair cosine +0.236, IsoScore 0.090), yet it still uses most of its dimensions: effective rank 728 of 1024. Its occupancy discrepancy sits at $z = -4,211$, and the whole-curve rank test rejects uniformity at the minimum attainable $p = 0.005$ with liftoff at $\cos \approx 0.82$: genuine tight-pair excess at the near-duplicate scale. The anisotropy-conditioned reference attributes most of that discrepancy to the cone itself.

The whole-curve test earns its place here. Reading each similarity level separately would put the liftoff at $\cos \approx +0.07$, a false alarm at the edge of the bulk of the kind §8 quantifies. Testing the curve as one object moves the significant excess up to the near-duplicate levels, where it belongs.

Operationally the corpus keeps 83% of the ideal noise budget: a query may drift to cosine

≈ 0.25 against its target before a wrong document is expected to win. The per-item field’s low tail is real but shallow, and the merge tree finds one modest pocket. Crowding here is therefore *diffuse within subsets* rather than pocketed. In that regime global averages mislead optimistically, while per-item counts still name the at-risk documents. The separability panel (labels provided, used only to condition the readout) reports two numbers that disagree. Neighborhood purity is 0.95: a document’s nearest neighbors nearly always come from its own subset. The silhouette is +0.04, near zero: the subsets occupy no regions of their own. The corpus is locally coherent yet globally entangled within the cone, and no single number could report both facts.

10 Governing Fine-Tuning with Measurements

The instrument finds crowding, locates it, and counts its cost in expected retrieval errors. This section tests a stronger claim: that its measurements can also *drive* interventions. The confusion law is differentiable, so the measurements can enter training in one of two roles — as the training *objective itself*, or as a *supplement* to an objective someone else has already chosen. Four uses are tested:

1. the measurements as the objective itself, on a worked example (below);
2. that same objective run as a gated loop on the case-study encoder (§10.3);
3. execution of the deduplication that the readouts prescribe (§10.5);
4. the measurements as a supplement to a supervised objective, in the form of guardrails (§10.6).

The two roles separate sharply, and the separation is the section’s main finding. Used as the objective itself, the measurements do not improve retrieval: the tuned encoder is retrieval-neutral against its base, and a label-supervised reference is far ahead of both. Used as a supplement — a *governor* over an objective someone else chose — they work well. Extra penalty terms built from the measurements halve the damage supervised fine-tuning does outside its training data at no cost to its gains, a gate rejects a defective round before any label is read, and the readouts route a repair that training could not perform. These measurements make a good governor and a poor objective, and §10.4 gives the reason.

10.1 Losses and the gradient-locality guarantee

For a batch $z_1, \dots, z_B$ (normalized) and a *measured* target scale σ :

$$\mathcal{L}_{\text{conf}} = \text{mean}_{(i,j) \notin \mathcal{E}} \Phi\left(-\frac{\|z_i - z_j\|}{2\sigma}\right), \quad \mathcal{L}_{\text{pres}} = \text{mean}_i \text{KL}\left(p_i^{\text{base}} \parallel p_i\right),$$

where $\mathcal{E}$ excludes positive and guarded pairs and p_i (p_i^{base}) are temperature-softmax similarity rows under the tuned (frozen base) model. $\mathcal{L}_{\text{conf}}$ is the batch estimate of $C(\sigma)$. Its gradient with respect to a pair distance falls off as $e^{-r^2/8\sigma^2}$ (Figure 12), so pairs farther apart than $\approx 3\sigma$ receive exponentially little of it. The objective therefore cannot disturb the arrangement of far-apart pairs, whatever the training schedule. The test suite enforces this, requiring a near

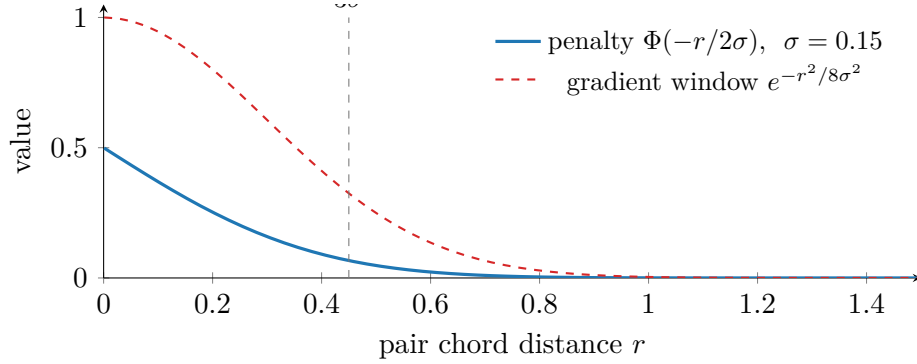


Figure 12: The confusion loss and its gradient window at a measured $\sigma = 0.15$ (the benchmark’s σ^*). The penalty falls away with distance, and the gradient falls away faster: beyond $\approx 3\sigma$ it is exponentially small.

pair to receive more than 10^3 times the gradient of an orthogonal pair. $\mathcal{L}_{\text{pres}}$ is the neighbor-overlap comparison in differentiable form — the explicit statement of “do not rewrite who is similar to whom.” (Implementation note: the chord computation floors $2 - 2c$ before the square root, because at exactly zero the root’s infinite derivative turns masked entries into $0 \cdot \infty$ NaN gradients.)

10.2 Mining, protocol, and measured effect

Batch composition is driven by the measurements in three ways.

- *Sampling*: items are drawn with weight proportional to their per-item collision counts, with a uniform floor so that the uncrowded bulk still anchors the geometry.
- *Negatives*: candidate negatives are drafted from the *confusable window* — the cosine range between the measured liftoff, where confusability begins, and the duplicate scale, where identity begins (operationalized as $\cos \approx 0.98$ throughout).
- *False-negative guard*: an anchor’s top- m base-model neighbors are never used as negatives, because the confusable window is precisely where unlabeled true relatives live.

The protocol is staged.

1. *Diagnose*, and take the cheapest fix that the readouts allow: a cone alone calls for a linear post-transform; duplicates call for data repair; only clustering that the cone cannot explain calls for training.
2. *Train* with $\mathcal{L}_{\text{task}} + \lambda_c \mathcal{L}_{\text{conf}} + \lambda_p \mathcal{L}_{\text{pres}}$ at the measured σ .
3. *Verify* on a held-out reservoir never seen by mining, and never grade with the statistic that was optimized.

Flagged items are the instrument’s crowded-item list: those with the highest expected collision counts under the base model. Their median expected collisions fall from 2.6 to

Table 10: The worked training example: a planted 300-item pocket in 4,000 items at $d = 64$, with a linear adapter standing in for the model and every verdict taken from a held-out slice unseen by mining or batching. The ceiling reached here belongs to the adapter rather than to the method: one linear map cannot unfold a tight pocket further without moving everything else, which is the trade $\mathcal{L}_{\text{pres}}$ enforces.

schedule	flagged median collisions	neighbor overlap w/ base
before	2.59	1.00
Adam, 10^{-2} , 600 steps, $\lambda_p=0.3$	1.25	0.92
Adam, 3×10^{-2} , 600 steps, $\lambda_p=0.3$	1.32	0.91
Adam, 3×10^{-2} , 1000 steps, $\lambda_p=0.5$	1.60	0.93

Table 11: The loop on the case-study corpus. Verdicts come from the held-out fifth of the 200,000-document measurement subset; the flagged cohort is the top 1% of items by base-model expected collisions, tracked at fixed $\sigma = 0.134$ across rounds. Rows are ordered by model capacity; the text reads the columns.

round	σ^*	liftoff	z	pocket (size)	overlap	cohort med.
base	0.1337	0.769	−3565	0.085 (294)	1.00	2.58
linear adapter	0.1331	0.770	−3665	0.064 (17)	0.979	2.82
LoRA, mismatched refs	0.1324	0.758	−3775	0.091 (319)	0.924	3.01
LoRA, corrected	0.1339	0.763	−3545	0.082 (307)	0.973	2.51
full fine-tune	0.1338	0.763	−3552	0.083 (275)	0.965	2.53

1.25, at a held-out neighbor overlap of 0.92. Raising the preservation weight trades recovery for overlap, as designed. The guard matters empirically: without it, window mining drafts unlabeled true relatives as negatives — the classic silent failure of hard-negative mining.

10.3 Closing the loop on a full-scale encoder

The synthetic example establishes the mechanism. What remains is whether the full loop — measure, build the objective from the measurements, train, re-embed, re-measure — operates on a contemporary public encoder and a million-document corpus, with the measurements adjudicating every round. We ran it on the case-study corpus (§9) and its encoder, Qwen3-Embedding-0.6B. A fixed, stratified 200,000-document measurement subset (proportional across the four subsets) was drawn once; 20% was held out and never touched by mining, sampling weights, or training. The objective is fully unsupervised. It combines $\mathcal{L}_{\text{conf}}$, evaluated at the subset’s measured $\sigma^* = 0.134$ under the false-negative guard, with $\lambda_p \mathcal{L}_{\text{pres}}$ against frozen base-model references. Batches are composed half from collision-weighted sampling and half from guarded pairs inside the confusable window of §10, which here runs from the liftoff at 0.769 up to 0.98. Between rounds, a rule fixed in advance — the *licensing rule* — decides whether to continue: σ^* must not fall, and held-out neighbor overlap must remain at least 0.90; that is, tuning may rewrite at most one held-out neighbor in ten.

Three findings follow from Table 11.

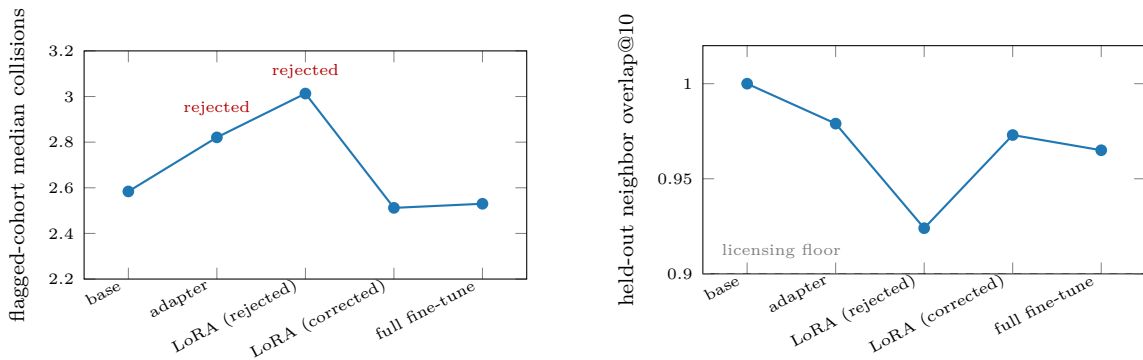


Figure 13: The encoder loop’s held-out verdicts by round (values from Table 11). Left: median expected collisions of the flagged cohort, tracked at fixed σ — the two rejected rounds *worsen* the cohort while their training losses decreased; the corrected rounds improve it. Right: neighbor overlap against the base model, with the pre-registered licensing floor; every accepted round stays above it.

The instrument rejects a misspecified round. The first LoRA round compared the tuned model against stored embeddings of full documents while the trainer embedded token-truncated views. The gradient therefore rewarded reproducing full-document embeddings from truncated text. The model was learning to imitate longer inputs rather than to separate crowded documents. Training losses fell throughout, while every held-out readout degraded: σ^* down, cohort collisions up 17%, the pocket enlarged. The licensing rule rejected the round from the held-out measurements alone, before any label was consulted.

Corrected training is safe but marginal. With references matched to the trainer’s own views and the confusable window present in every batch, both LoRA and full fine-tuning leave every readout intact: held-out overlap stays at 0.97 and the flagged cohort improves by 2–3% in the median. The effect is small but nearly universal across items. Of the 408 flagged items in the held-out split, 406 under LoRA and 356 under the full fine-tune individually reduce their expected collisions (paired sign tests, $p < 10^{-100}$ and $p < 10^{-56}$). σ^* is unchanged rather than merely noisy: the paired difference across eight independent pair samples is +0.00007 against a per-sample standard deviation of 0.00012. This holds even though the full fine-tune moves roughly 40× the parameters of the LoRA round on a doubled pair batch.

Capacity is not the binding constraint. The linear adapter halved collisions on the synthetic single-pocket benchmark (Table 10), yet the real corpus rejects it: σ^* falls and cohort collisions worsen by 9%, because one linear map cannot serve the corpus’s mixture of scales. LoRA and the full fine-tune land within noise of one another. The corpus’s dominant pocket is textual near-duplication, and the preservation term refuses to separate items whose inputs are near-identical. This confirms the staged protocol’s routing rule from the training side: duplicate pathology is a data-repair problem, and no objective that preserves base-model semantics can substitute for deduplication. Training at the measured scale is safe, but it is not a shortcut around the data.

As a final check that the subset verdicts transfer, the full corpus was re-embedded with

the accepted fine-tune and measured at full scale with the identical pipeline as the base case study. Every readout reproduces within measurement noise: σ^* 0.1227 against the base 0.1226; liftoff 0.817; effective rank 728; IsoScore 0.089; mean pair cosine 0.235; hubness skew 1.88; top pocket 198 items at prominence 0.090 against the base’s 204 at 0.090. Two things follow. The accepted fine-tune preserves the geometry at 10^6 documents, and the 200k held-out measurement is a faithful stand-in for the full corpus.

Reading Figure 13. The left panel shows the counterfactual that the instrument prevented. Both rejected rounds sit *above* the baseline — their tuned models made the flagged cohort worse — while their training losses fell monotonically. Loss alone endorsed those rounds; only the held-out geometric readouts saw the regression. The right panel shows the gate that caught them: the rejected LoRA round is the only point below the overlap floor fixed in advance, and the accepted rounds cluster near the base model’s neighborhood structure. The contribution here is not the marginal improvement of the accepted rounds. It is that on every round the labels later scored, the held-out geometric verdict agreed in sign with the label-based evaluation, at no label cost.

10.4 What the objective alone does to retrieval

The loop above never touches a relevance judgment. Judgments enter once, after every round is frozen, and only to score predictions that the instrument had already committed to. On the corpus’s citation-grounded question-to-section retrieval set (1,000 held-out questions against a 35,173-section sub-corpus, exact search), the unsupervised full fine-tune moves recall@1 from 0.288 to 0.301 and MRR@10 from 0.467 to 0.475; the LoRA round is flat (0.286, 0.466). The recall gain does not reach significance at this query count: McNemar’s exact test on the paired outcomes gives $p = 0.45$ (118 against 131 discordant queries), and the bootstrap interval on the MRR@10 difference, $[-0.008, +0.023]$, spans zero. The honest reading is therefore neutral to positive: the geometry improves item by item, while the aggregate task effect is bounded near zero. That is consistent with the duplication finding, since most failures here are not caused by the crowding this objective is permitted to repair.

For context, a label-supervised fine-tune measured under the same harness reaches recall@1 0.355, an upper reference point supplied by supervision rather than by geometry.

The per-item layer makes a sharper, item-level prediction: documents with high base-model expected collisions should be the ones that *fail at their own queries* (a judged-relevant document outranked by non-relevant neighbors). Scored blind, base expected-collision counts predict which documents fail at AUC 0.578, against 0.551 for the natural control (top-1 neighbor cosine), over 953 gold documents. The signal is better than chance (bootstrap 95% interval $[0.536, 0.622]$), though the margin over the proximity control is not individually significant on this single corpus ($+0.028$, paired bootstrap interval $[-0.017, +0.073]$). §11 settles that comparison across five corpora and seven encoders.

This section has exercised the whole map from readout to intervention:

- a cone and nothing more calls for a linear post-transform;
- named near-duplicate pockets route to data repair before any training;
- clumping that the cone cannot explain licenses guarded training at the measured σ^* , checked on held-out measurements against a rule fixed in advance;

Table 12: Dedup routing scored blind on 1,000 frozen queries (fixed/broken = queries whose failure status at rank 10 changed). Confusable-window cleanup applies to foreign distractors; only true-duplicate merging applies within the formulaic corpus.

threshold	foreign distractors removed	target corpus merged
liftoff (0.82)	12 fixed, 0 broken ($p=0.0005$)	26 fixed, 57 broken ($p=0.0009$)
duplicates (0.95)	no change	9 fixed, 2 broken ($p=0.065$)

- per-item collision counts rank which documents to repair or re-chunk first;
- σ^* sets the query-noise budget an application may spend.

Every entry on that map is a governing action: it decides what to do, where, and at which similarity level, or it vetoes a step someone else proposed. None of them is the objective doing the work by itself. That distinction is the section’s result, and it follows from the mechanism above rather than from a shortfall of capacity or tuning. A crowding measurement can only be trained away when the crowding is something training is permitted to move, and the near-duplicate mass that dominates this corpus is not. The external grid shows the same limit from the other direction: failures caused by composition rather than by crowding are invisible to the per-item field (§11). Used to govern — to select the repair, to bound the damage of a supervised objective, to reject a round — the measurements pay for themselves; used as the objective, they do not.

10.5 Executing the routing rule

We tested the routing claim directly, executing the repair that the measurement prescribed while blind to all labels, then scoring on the frozen evaluation. The repair has three mechanical steps:

1. identify near-duplicate groups purely from geometry (pairs above a stated cosine threshold, grouped by connectivity);
2. reduce each group to one canonical member, chosen by identifier order;
3. map relevance judgments through that reduction mechanically.

Two repair sites and two scales give four combinations, and only one of them is a repair (Table 12).

Removing *foreign* near-duplicates inside the confusable window — the band that opens at the liftoff cosine — fixes 12 queries and breaks none (McNemar exact $p = 5 \times 10^{-4}$). This is the retrieval gain that training could not deliver. Merging *within* the formulaic target corpus at that same scale is harmful, breaking 57 queries: these regulatory sections share boilerplate, so their confusable window holds legally distinct material. Within the corpus, only merging at the true-duplicate scale ($\cos \geq 0.95$) helps.

The two scales the instrument reports are therefore not interchangeable. Liftoff marks the onset of confusability rather than the duplicate boundary, and each scale licenses a different repair — a mapping the held-out labels confirm in both directions (Figure 14).

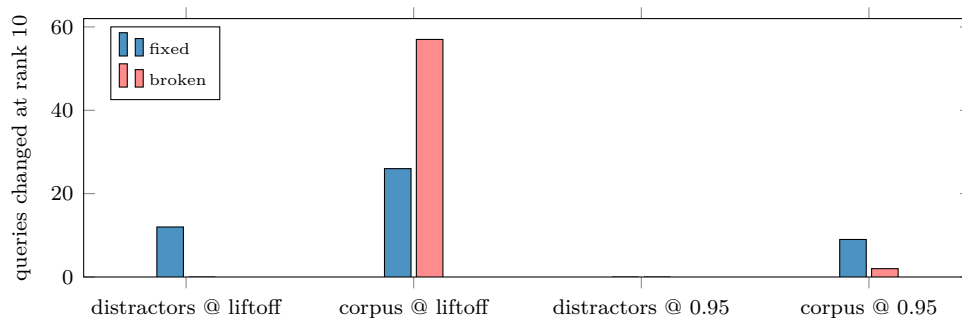


Figure 14: The routing dissociation as counts (values from Table 12). At the liftoff scale, removing *foreign* confusable material only fixes; merging *within* the formulaic corpus at that same scale mostly breaks. At the true-duplicate scale the pattern reverses in both sites. Each repair is licensed at exactly one scale and one site — the mapping the instrument’s readouts prescribe.

Reading Figure 14. “Deduplicate the corpus” names a single operation, but its effect depends on site and scale: of the four cells in the figure, only one is cleanly positive. The measurement supplied the two numbers that disambiguate the operation — the liftoff cosine, where confusion begins, and the duplicate scale, where identity begins — and each licenses a different intervention.

10.6 Guardrails for supervised fine-tuning

This is the second role: the measurements supplement a label-supervised objective rather than replacing it. The motivation is a specific failure. A prior supervised fine-tune of the case-study encoder improved first-rank retrieval on the task it was trained for, while quietly collapsing a multi-hop evaluation it was not trained for. That is the failure class $\mathcal{L}_{\text{pres}}$ and the false-negative guard exist to prevent. We ran the identical supervised recipe (in-batch contrastive loss on 45,465 question–section pairs, two epochs) in two arms differing only in the guardrails.

- *Control*: the task loss alone.
- *Guarded*: the task loss plus $\lambda_p \mathcal{L}_{\text{pres}}$ against the frozen base on identical truncated views. In addition, any batch document whose base-model cosine to the anchor’s positive exceeds the measured liftoff is masked out of the contrastive denominator, so the task loss never pushes apart pairs that the base model considers confusable relatives.

The gate ranked the arms before any label was read. The guarded model leads on every label-free readout: neighbor overlap 0.650 against 0.537, σ^* 0.146 against 0.136 (the base being 0.134), and flagged-cohort collisions 0.47 against 2.10.

The frozen evaluations confirmed the prediction. On the targeted task the guardrails cost nothing: MRR@10 is 0.503 against 0.501, both far above the base 0.467. On the untargeted open evaluation the guarded arm wins every metric, with single-document MRR@10

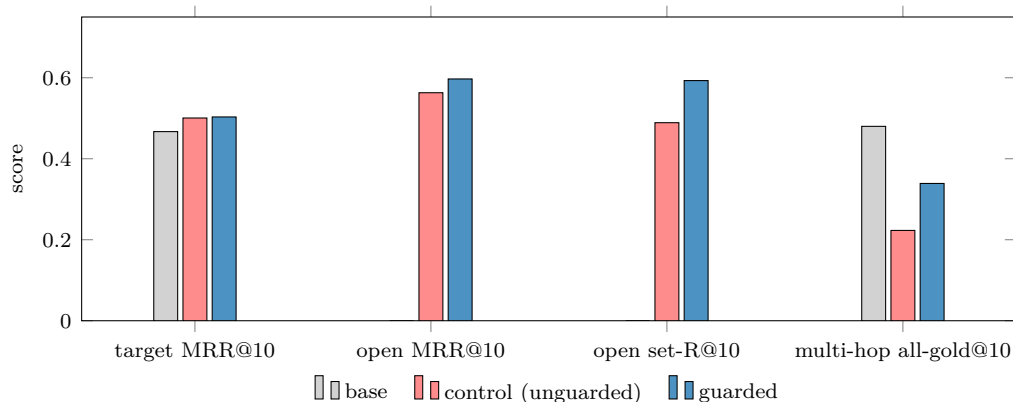


Figure 15: The guardrail experiment on the frozen evaluations, four metric groups left to right (set-R@10 is the recall of the full judged set within the top ten). Base bars for the two open metrics were not measured under this harness and are omitted; the multi-hop base bar (≈ 0.48) is the earlier baseline. The text reads the comparison.

at 0.597 against 0.563. The decisive metric is all-gold@10, the fraction of multi-hop queries whose every judged-relevant document reaches the top ten, because that is where the earlier regression appeared. There the guarded arm scores 0.339 against the control’s 0.223 — a 52% relative improvement over 1,496 held-out queries, a gap exceeding five standard errors. The control reproduces the earlier collapse, while the guard halves the damage and improves every other number.

Two conclusions follow. Measurement-derived guardrails make supervised domain adaptation safer at no cost to its gains, and the label-free held-out gate selects the better model before any label is consulted.

Reading Figure 15. Compare the control and guarded bars group by group, left to right. On the targeted metric the two are indistinguishable: whatever the guardrails withheld from the training signal, none of it was needed for the supervised gain. The three open-evaluation groups then separate progressively, widest on the multi-hop metric. That is the ordering to expect if the control’s damage falls on *relations among documents* rather than on query–document matching, since the guard’s masked pairs and the preservation term are both relational. The label-free gate read before scoring shows the same thing: the control had rewritten 46% of held-out neighborhoods (overlap 0.537) against the guarded arm’s 35%, and multi-hop retrieval depends on exactly those neighbor identities.

11 Results Across Corpora and Encoders

The evidence so far comes from synthetic benchmarks and one real corpus. This section tests the instrument on five judged public collections and seven public encoders (Tables 1 and 2). Each combination of a corpus and an encoder is one *cell* of the grid, treated as its own embedded corpus — first measured, then scored. Five corpora under seven encoders give 35 cells, plus the legal case study. Each cell follows four steps in strict order:

1. embed the corpus;

Table 13: The external grid at a glance (7 encoders per corpus; readouts frozen before judgments). ρ compares two rankings of the seven encoders within one corpus — by σ^* and by nDCG@10 — as a Spearman rank correlation; “control” repeats this with mean pair cosine in place of σ^* . The final columns give the range across encoders of the per-document score’s failure-prediction AUC (a failure: a judged-relevant document ranked past 10 at its own queries) and how many encoders score above chance.

corpus	task type	$\rho(\sigma^*)$	control	failure AUC	above chance
Quora	duplicate matching	+0.75	−0.79	0.66–0.74	7 of 7
ESCI	product search	−0.07	+0.32	0.57–0.69	7 of 7
Natural Questions	open-domain QA	−0.54	+0.54	0.51–0.62	7 of 7
HotpotQA	multi-hop QA	−0.75	+0.71	0.48–0.59	4 of 7
TREC-COVID	biomedical	−0.32	+0.39	(saturated)	—
legal (case study)	—	—	—	0.58	1 of 1

2. compute the full readout set and record its hash, *before* any judgment file is opened;
3. compute the per-document collision field of §7 (each document’s expected collision count) at the cell’s measured σ^* ;
4. only then run and score the queries (nDCG@10, recall@100, and per-relevant-document failure labels).

As a pipeline check, BGE-large’s score on TREC-COVID reproduces its published benchmark value. Every cell appears in Table 13 and in the figures that follow it.

Reading Figures 16–17. In the scatter row, the claim under test is a slope: if geometric headroom explained retrieval quality, every panel would rise to the right. Only the duplicate-matching panel rises. There, the conventional summary (mean pair cosine) points the *wrong* way: in the one setting where geometry is the task, the popular statistic misleads. The AUC strip asks the same question inside each corpus. There one encoder and one corpus are fixed, so semantic alignment cannot vary. The geometric signal is positive in every informative cell except the multi-hop group, drawn in red. Together the two views separate two questions. Across encoders, retrieval quality has two ingredients: the geometry, and whether the encoder places related texts near each other. Geometry predicts only where the second ingredient is roughly constant. Within a single embedding the second ingredient cannot vary, so geometry predicts failure wherever the failure comes from items being hard to tell apart — here, in every informative cell except the multi-hop ones. The prominence strip makes the same point for structure. Whether a corpus forms pockets depends on the corpus–encoder *pair*: the spread within a column is as large as the spread across columns. That is why the instrument reports structure per embedding rather than labeling corpora as clean or dirty.

The per-item claim holds on four corpora and fails on the fifth. We call a cell *informative* when its failure label varies: some judged documents fail at rank 10 and some do not. A label that is nearly constant cannot be predicted by anything. The informative cells comprise all seven encoders on each of four corpora — duplicate questions, product search, open-domain QA, and multi-hop QA — plus the legal case study, giving 29 in all. In 26 of them, expected-collision counts predict which judged-relevant documents fail at their own queries (sign test $p \approx 8 \times 10^{-6}$); the per-corpus AUC ranges are the final columns of

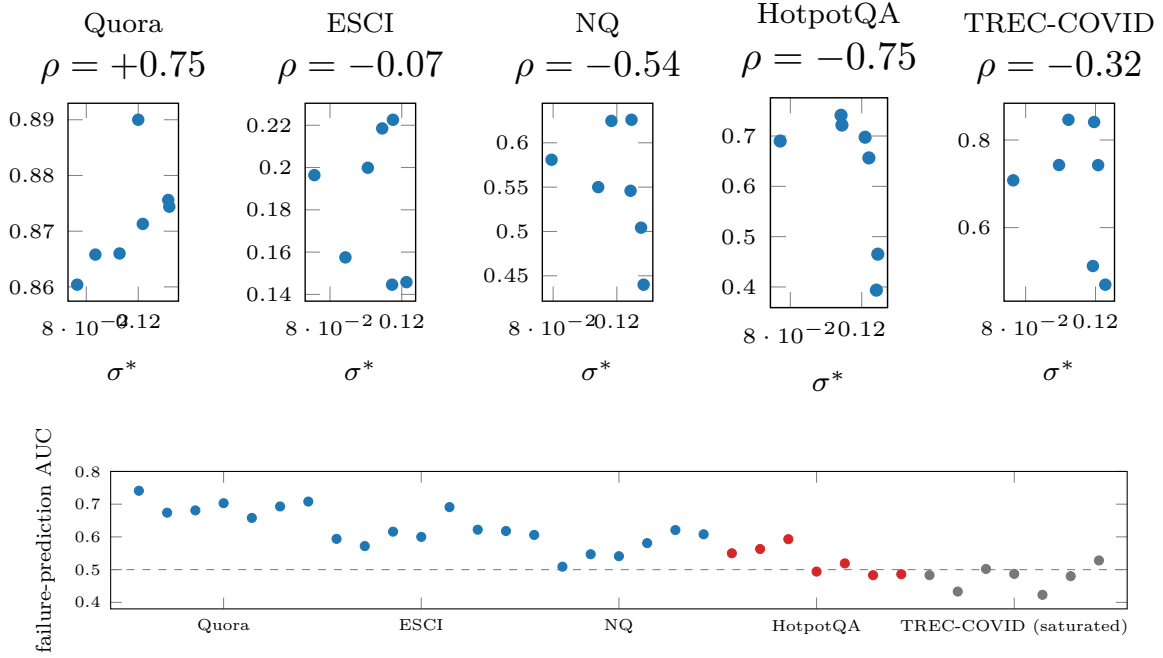


Figure 16: Every grid cell, plotted. Top: within each corpus, the seven encoders’ σ^* (horizontal) against their nDCG@10 (vertical), with each panel’s rank correlation printed above it. Bottom: the per-document collision field’s failure-prediction AUC for all 35 cells, grouped by corpus, against the chance line. It sits above 0.5 in every informative cell outside the multi-hop group (red), where three of seven encoders fall to chance; the TREC-COVID group (gray) is uninformative because its outcome label saturates (98% of judged documents past rank 10).

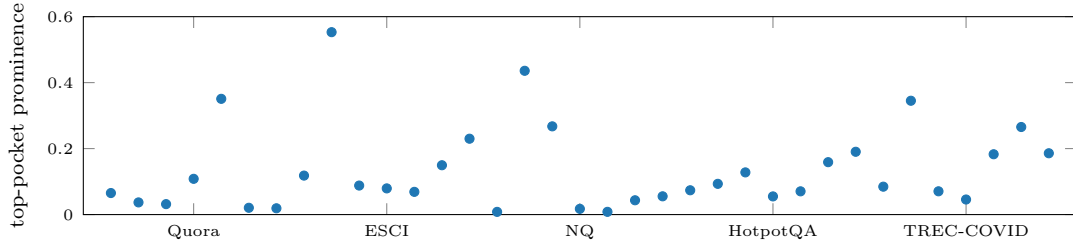


Figure 17: Structure across every corpus–encoder pairing: the most prominent pocket’s merge-tree prominence for every corpus–encoder cell (all 35 exceed the null-calibrated floor). The spread within each corpus is as large as the spread across corpora: encoders disagree about which corpora form tight substructure. The text draws the consequence.

Table 13. Duplicate questions, product search, and open-domain QA are positive in every cell, including an AUC of 0.69 on product search for the case-study encoder despite an 84% base failure rate. The biomedical collection is excluded as uninformative by construction rather than by counterexample: with 50 queries carrying hundreds of relevant documents

each, 98% of judged documents exceed rank 10 and the failure label saturates entirely.

Multi-hop retrieval is where the per-item claim stops. On HotpotQA the field predicts failure for four of seven encoders (0.55–0.59) and sits at chance for the other three (0.483, 0.486, 0.494). These cells are not saturated — their failure rates run from 23% to 58%, the most balanced labels anywhere in the evaluation — so the null result is a real one. The explanation is simple. Embeddings organize documents by *similarity*. Multi-hop questions retrieve by *relationship*: the answer lies in how two documents connect, and in neither one’s neighborhood. A document can therefore fail at its own query while sitting in uncrowded space, and reducing similarity-crowding cannot fix a relationship-based failure. This is the per-item counterpart of the corpus-level scope law below. Both say the same thing: the instrument measures whether items remain distinguishable, and that predicts a retrieval failure only when the failure is caused by items being hard to tell apart. Where relevance requires composition or reasoning, the measurement is silent, and reporting it as silent is more useful than reporting a number.

The corpus-level claim earns only a narrow scope. Within one corpus, across the seven encoders, the rank correlation of σ^* with nDCG@10 reaches +0.75 only on duplicate matching, the one task where relevance *is* geometric proximity. On that same task the mean-cosine control is *inverted* (−0.79). On product search the correlation is near zero (−0.07): shopping relevance mixes attribute matching with intent that the geometry cannot see. On open-domain QA, multi-hop QA, and biomedical search it reverses (−0.54, −0.75, −0.32): the smallest encoder has the best geometry and the worst retrieval. The two knowledge-seeking corpora agree, which is what distinguishes a scope law from a single anomalous corpus. The mechanism is the alignment–uniformity decomposition. Retrieval quality has two ingredients: semantic alignment (whether the encoder places related texts near each other) and geometric headroom (whether the space leaves room to tell neighbors apart). **ambit** is unsupervised by construction, so it measures only the second. σ^* therefore ranks encoders only where alignment is essentially constant across them, and a semantically weak encoder can attain excellent geometry by spreading apart documents it does not understand. The operational conclusion is that corpus-level σ^* is a budget for one corpus rather than a ranking across encoders. It compares one corpus against itself under encoder variants, repair, or drift, which is how the guardrail gate used it. For comparison across encoders, the validated readout is the per-item collision field above.

The two similarity levels, checked against annotated duplicates. The routing experiment (§10.5) rests on there being two distinct similarity levels — where confusability begins, and where true duplication begins. The duplicate-question collection annotates true duplicates, so this can be checked directly: predicting “duplicate” from pair cosine alone becomes precise only at the higher level (0.29 precision at $\cos = 0.80$ against 0.93 at 0.95), exactly as the two-level reading requires (Figure 18).

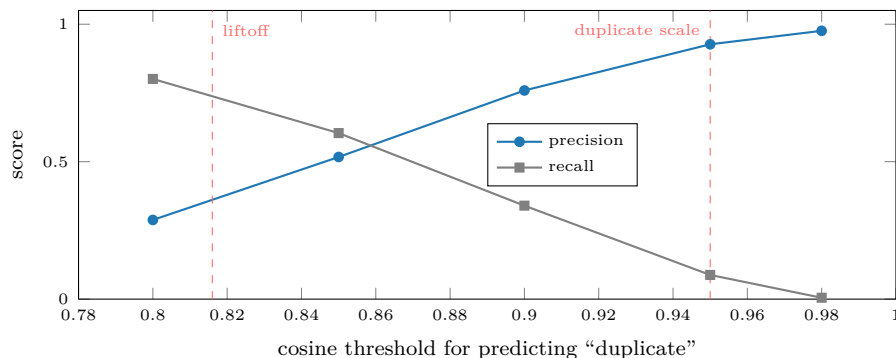


Figure 18: Duplicate prediction from pair cosine alone on Quora’s annotated duplicate pairs (case-study encoder; 15,675 gold pairs). Precision rises monotonically toward the duplicate scale while recall falls; the dashed lines mark the two scales — liftoff and duplicate boundary — whose contrast the text interprets.

12 Limitations and Future Work

- **Breadth of evidence.** Five collections and seven encoders were enough to expose the boundaries reported in §11; more collections, deliberately degraded encoders, and independent replication would harden them. The multi-hop result also makes a cheap, testable prediction: the per-document score should recover on multi-hop questions whose answer lies in a single document, and stay at chance on the rest.
- **The objective, retested where it should work.** Training at the measured σ^* has been tried on one corpus. That corpus’s crowding is textual duplication, which a semantics-preserving objective must leave in place, and retrieval was unchanged. The prediction that follows is testable and cheap: on a corpus whose failures really are caused by items being hard to tell apart — duplicate matching, where the per-item field reaches an AUC of 0.74 — the same objective should produce a measurable retrieval gain. If it does not, the objective should be withdrawn as a contribution and the training layer stated purely as a governor.
- **One untested prediction.** The exact confusion transform and its near-equivalent alternative order corpora almost identically, but they disagree on one case (Table 6): the exact form ranks duplicate pathology above diffuse collapse. That disagreement is falsifiable and has not been tested at scale.
- **Query distributions.** The instrument currently treats the corpus’s own documents as stand-ins for queries. Accepting an optional *unlabeled* set of real queries would replace that assumption with a measured confusion between queries and documents, still without using labels.
- **Sharper aggregates.** Adding up the per-competitor error probabilities (a *union bound*) can only overstate the true collision count, never understate it — but it overstates most where a few documents sit near very many others. Gaussian comparison inequalities may give bounds from both sides.

- **Recoverability.** A useful next readout is how much of the measured crowding a limited class of transforms could remove. Computing that fraction, rather than predicting it, would separate crowding caused by the embedding model from crowding inherent in the data.

13 Conclusion

Crowding is a failure that embedded systems exhibit without raising any error. Global averages cannot see it, and binned displays cannot be trusted to measure it. The instrument described here rests on one commitment applied uniformly: read continuous quantities at every similarity level, against stated references, and name the individual documents responsible.

The mathematics is classical — Ripley’s curve, Stolarsky’s identity, the distance to a measure, the condensed cluster tree, a Gaussian tail bound. The contribution is their selection, calibration, and composition into a system with well-understood properties, traced end to end through one benchmark whose planted pathology every layer detects, localizes, converts into an expected error count, and partially repairs. Beyond that benchmark, the per-document score identified the documents that later failed at their own queries, in 26 of 29 testable pairings; the three exceptions are multi-hop questions, where composition rather than crowding causes the failure. The measurements also chose interventions that held up once the labels were read. A deduplication fixed twelve queries and broke none. Penalty terms halved the damage a supervised fine-tune did outside its training data. A veto stopped a training round that its own falling loss endorsed. Each of those governs a decision that someone else’s objective would otherwise make alone. Run as the training objective itself, the same measurements left retrieval unchanged. That negative result is stated at the same prominence as the successes, because an instrument’s reliability rests in part on a precise statement of its limits.

Appendix A Engineering Notes

The implementation is organized so that its asymptotic costs match the claims of §9: one streaming pass over the corpus, followed by statistics whose cost is independent of corpus size. This appendix records the implementation decisions behind the measured constants of Table 8, and collects the default configuration referenced in §4:

setting	default
pair sample	2×10^5 sampled distinct pairs
whole-curve test	rank envelope over 199 null curves (99 in figures)
anisotropy reference	sampled from the corpus’s own covariance eigenvalues
per-document field	distance-to-measure mass fraction $m = 0.02$
tight groups	minimum group size 8
discrepancy z	64 null replicates

Streaming scan. Ingestion is a single sequential pass over raw `.npy` files (memory-mapped) or sharded Parquet (chunked decode), in batches of 5×10^4 rows. Each batch of b rows updates the running mean, the norm moments, and the $d \times d$ second-moment scatter through one $X_b^\top X_b$ GEMM, and feeds a bounded reservoir sample. Peak memory is therefore $O(d^2 + \text{reservoir})$ whatever the corpus size. The covariance spectrum is extracted once, at the end of the pass. An optional approximate mode terminates the pass after a prescribed number of rows and estimates the spectrum from that prefix; §9 documents the failure mode this introduces on sorted shards.

Blocked kernels at reservoir scale. No computation materializes an $m \times m$ matrix for reservoir size m . Top- k cosine values (the input to the distance-to-measure field) and collision counts are computed from $b \times m$ Gram tiles with partial-selection top- k (`argpartition`, linear per row, in place of a full sort). The mutual-reachability minimum spanning tree uses a vectorized Prim iteration that maintains a frontier-distance array: $O(m^2)$ arithmetic executed as full-array operations, $O(m)$ memory beyond the current tile. The structural layer additionally caps its inputs (6,000 points for the DTM field, 4,096 for the merge tree), which fixes the corpus-size-independent constants observed in Table 8.

Null sampling without vector synthesis. All isotropic-null quantities exploit the closed form for the pair cosine of independent uniform directions on S^{d-1} , $c = 2B - 1$ with $B \sim \text{Beta}(\frac{d-1}{2}, \frac{d-1}{2})$: null pair samples are drawn directly from this scalar law. A 199-replicate rank envelope therefore requires 199 scalar-array draws rather than 199 synthetic corpora. The anisotropy-conditioned reference uses the rotation invariance of the pair-cosine law under the angular central Gaussian — only the covariance spectrum enters — so reference pairs are generated from a small pool of spectrum-weighted Gaussian directions with no eigenvector computation. The Gaussian confusion law is evaluated with a vectorized rational approximation to erf (absolute error below 1.5×10^{-7}), which keeps the core free of dependencies beyond `numpy`.

One shared pair sample. A single random-pair cosine sample (default 2×10^5 pairs) is drawn once per corpus and reused by every pairwise statistic — $K(t)$, the envelope test, the

discrepancy z , the collision count $C(\sigma)$, and σ^* . The continuous layer therefore costs one pass over a fixed-size array per statistic, and all reported statistics refer to the same measured sample.

Distributed training and pair-batch statistics. The encoder-loop trainer (§10.3) runs data-parallel: each rank embeds its own rows and the embeddings are gathered across ranks (the local slice retaining its gradient) before the pair losses, so pair statistics scale with world size at fixed per-device memory. This matters beyond throughput. The pair losses see only within-batch pairs, so small pair batches permit batch-level overfitting: the model separates the pairs it is shown while degrading structure it never sees. The loop’s held-out measurements catch that failure. Gathered batches must be identically shaped on every rank; the batch builder emits a fixed count of distinct rows per rank for this reason.

Optional acceleration tiers. The core is `numpy`-only, with the heavy kernels routed through multi-threaded BLAS. An optional module provides the same three kernels — covariance accumulation, pair cosines, and exact blocked k NN — on `torch` devices (CPU, CUDA, or Apple `mps`, with automatic selection), transferring each chunk to the device as it streams. An approximate-neighbor path through FAISS is available where installed. These tiers change constants rather than asymptotics; every number reported in this paper was produced by the `numpy`-only configuration (Appendix Appendix C).

Appendix B The Report Artifact

The toolkit delivers its measurements as a single self-contained, theme-adaptive HTML report that makes no external requests. Its design is documented here for reproducibility. The artifact is engineering rather than an academic contribution: the contribution of this report is the measurement layer of §6–§7, exposed as library functions independent of any rendering.

Two design rules follow directly from the measurement theory. *Reference curves are comparisons only*: every panel draws its reference curve, because absolute cosine is not self-interpreting (§2). *Display versus measurement*: two-dimensional projections lay figures out but never carry a statistic, because projected density is not corpus density (§5).

The remaining choices are presentation craft. One accent color is used, and meaning is carried by position along annotated axes. Figures are ordered so that each answers the question the previous one raises: whether crowding exists, who is affected, and what structure they form. Every panel carries a short explainer stating the context a reader would otherwise have to assume — how the null was built, the subsample sizes, and where the identifiers come from. One legibility finding is worth recording, because it reinforces the display/measurement separation. A minimum-spanning-tree view drawn over the projection stayed too busy to read on real corpora, even when restricted to the tightest bridges, and became a hidden option. The condensed tree of §6.4 shows the same object legibly, as prominence bars.

Appendix C Test Environments

The hardware is incidental to the results, and only three facts about it bear on a claim in this report.

- Every reported *measurement* was computed by the released `numpy` core, on Python 3.13 with multi-threaded OpenBLAS. That is the configuration the portability claim is about.
- Embedding, fine-tuning, and the exact per-document collision fields at million-row scale ran on separate accelerated hardware — a workstation with four GPUs of 96 GB each — through the artifact’s optional `torch` path. Those steps produce inputs to measurements, never measurements themselves.
- No experiment uses code outside the artifact and its stated experiment scripts, and random seeds are fixed and stated per experiment.

Appendix D Formula Glossary

Every formula in the report is restated here in plain language: what it formulates, why it matters, how it is used, and what every symbol and operation means. Entries follow the order of first appearance. The first two entries explain the shared notation and the recurring vocabulary; each later entry still lists all of its own symbols.

How to read the notation (used throughout).

- n — the number of documents in the corpus. d — the number of coordinates in each embedding vector (the “dimension”).
- i, j — indices naming individual documents, each running from 1 to n . Writing x_i means “the embedding vector of document number i ”; a pair (i, j) names two different documents.
- $\Pr[\text{event}]$ — “the probability that the statement inside the brackets is true.” A subscript names what is random: in $\Pr_{(i,j)}[\dots]$ the random thing is the choice of pair — draw two distinct documents at random, and ask how often the bracketed statement holds. With no subscript, the randomness is stated in the entry (for the flip law it is the query noise).
- $\{\sigma : C(\sigma) \leq 1\}$ — “the set of all values σ such that the condition after the colon holds”; $\max\{\dots\}$ takes the largest member of a set.
- $\sum_{j \neq i}$ — add up the expression that follows, once for every allowed value of the index (j running over all documents except i). $\frac{1}{n^2} \sum_{i,j}$ — add over all n^2 ordered pairs, then divide by their count: an average over pairs.
- $\int f(r) dK(r)$ — an average of f weighted by how often each distance r occurs among pairs; “ $dK(r)$ ” means “the fraction of pairs whose distance is (approximately) r .”
- $\mathbb{E}$ — the expected value (average) over the stated randomness.
- $\log$ — the natural logarithm; e^x — the exponential function, its inverse.

- $\partial f / \partial r$ — the derivative of f with respect to r : how fast f changes when r changes a little. $\propto$ — “proportional to”: equal up to a constant factor that does not depend on the variable under discussion.
- $\|v\|$ — the Euclidean length of a vector v (square root of the sum of its squared coordinates). $\langle u, v \rangle$ — the dot product of two vectors (multiply matching coordinates, add them up).
- Φ — the standard normal cumulative distribution function: $\Phi(u)$ is the probability that a standard Gaussian (bell-curve) random number falls below u . $\Phi(0) = 1/2$; $\Phi(-4) \approx 3 \times 10^{-5}$.

Words used throughout (not formulas).

- *similarity level* — a value of the cosine at which the corpus is examined, $\cos = 0.9$ for instance. “At every similarity level” means a statistic is read at all of them rather than at one chosen cutoff.
- *null model* — a simulated corpus of the same size and dimension used as the comparison; no real corpus matches it, and the distance from it is what the statistics report. Two appear in this report: points spread as evenly as chance allows, and the same with the corpus’s cone reproduced.
- *region* (also *cap*) — a patch of the sphere: every direction lying within some angle of a chosen centre.
- *pair share* — the fraction of all document pairs falling inside a group. A group holding a fraction f of the documents holds about f^2 of the pairs, which is why a small group is diluted in any average over pairs.
- *pocket* — a group of documents sitting tightly enough together to be reported against the null floor.
- *guard terms* — extra penalty terms built from the measurements and added to someone else’s training objective, rather than replacing it.
- *flagged cohort* — the documents whose base-model collision counts mark them as most crowded (the top 1%), tracked across training rounds to see whether an intervention helped them.
- *cell* — one corpus embedded by one encoder: the unit of the multi-corpus evaluation (§11).

Unit vectors and cosine similarity: $c_{ij} = \langle x_i, x_j \rangle$, **points on $\mathbb{S}^{d-1}$.**

- **What it says:** Every document is represented as a length-one arrow in d -dimensional space, and the similarity of two documents is the cosine of the angle between their arrows.

- **Why it matters:** Cosine is the score retrieval systems actually rank by, so all geometry in the report is expressed in the same units the system uses.
- **How it is used:** Every statistic in the report is a function of these pairwise cosines.
- **Term by term:**
 - x_i — the embedding vector of document i , rescaled to length 1 (each of its d coordinates is a number; the whole vector is one point in d -dimensional space).
 - $\mathbb{S}^{d-1}$ — the unit sphere in d dimensions: the set of all vectors of length exactly 1. The superscript $d - 1$ is the standard name for the sphere’s surface (a globe in 3-D space has a 2-D surface); nothing in the report depends on this convention.
 - $\langle x_i, x_j \rangle$ — the dot product of the two vectors, which for length-one vectors equals the cosine of the angle between them.
 - c_{ij} — shorthand for that cosine between documents i and j : +1 means identical direction, 0 unrelated (perpendicular), -1 opposite.

Chord distance: $r_{ij} = \|x_i - x_j\| = \sqrt{2 - 2c_{ij}}$.

- **What it says:** The straight-line distance between two points on the sphere, computed directly from their cosine.
- **Why it matters:** The query-noise law below is exact in chord distance rather than in cosine, so the report converts between the two constantly; this identity is the dictionary.
- **Term by term:**
 - $x_i - x_j$ — the vector pointing from document j ’s position to document i ’s.
 - $\|x_i - x_j\|$ — that vector’s length: the ordinary straight-line distance between the two points.
 - r_{ij} — shorthand for that distance.
 - $\sqrt{2 - 2c_{ij}}$ — the same distance computed from the cosine alone. It follows by expanding the squared length: $\|x_i - x_j\|^2 = \|x_i\|^2 + \|x_j\|^2 - 2\langle x_i, x_j \rangle = 1 + 1 - 2c_{ij}$.
 - Reference points: identical items have $r = 0$; perpendicular ones $r = \sqrt{2} \approx 1.41$; opposite ones $r = 2$.

The null cosine scale: $\text{coordinates} \sim \mathcal{N}(0, 1/d)$, **typical cosine** $\approx 1/\sqrt{d}$.

- **What it says:** If points are spread uniformly over a high-dimensional sphere, each coordinate behaves like a Gaussian random number with variance $1/d$, and the cosine between two random points is almost always within a few multiples of $1/\sqrt{d}$ of zero (at $d = 768$, about ± 0.036).
- **Why it matters:** This is the baseline against which “suspiciously similar” is judged: in high dimension, unrelated items are *very* nearly perpendicular, so even modest cosines are extremely surprising under the null.

- **Term by term:**

- $\sim$ — “is distributed as”: the quantity on the left behaves like a random draw from the distribution on the right.
- $\mathcal{N}(0, 1/d)$ — the normal (bell-curve) distribution with mean 0 and variance $1/d$: values concentrate near zero, spread $\approx 1/\sqrt{d}$.
- $1/\sqrt{d}$ — the typical size of a null cosine; it shrinks as dimension grows, which is the concentration phenomenon.

Pair-closeness curve: $K(t) = \Pr_{(i,j)}[c_{ij} \geq t]$.

- **What it says:** For every similarity threshold t simultaneously, the fraction of document pairs that are at least t -similar.
- **Why it matters:** It is the report’s primary object: a complete, bin-free record of how much of the corpus sits at every closeness scale. Crowding appears as an excess of very close pairs — the curve “lifting off” above its reference at high t .
- **How it is used:** Estimated exactly by sorting a large random sample of pair cosines; compared against the two null curves below; the comparison is tested with the rank-envelope test.
- **Term by term:**
 - K — the name of the curve (after Ripley’s K function in spatial statistics, of which this is the spherical, cosine-coordinate version). For each input t it returns one number between 0 and 1.
 - t — a similarity threshold, swept continuously from low to high; the curve is read at every t at once, which is what “no bins” means.
 - (i, j) — a pair of two distinct documents, chosen uniformly at random from the corpus.
 - $\Pr_{(i,j)}[c_{ij} \geq t]$ — the probability, over that random choice of pair, that the pair’s cosine reaches the threshold — equivalently, simply the fraction of all pairs that do.

The uniform null for pair cosines: $c = 2B - 1$ with $B \sim \text{Beta}(\frac{d-1}{2}, \frac{d-1}{2})$.

- **What it says:** If points were spread perfectly evenly on the sphere, the cosine of a random pair would follow this exact, known distribution — a Beta distribution stretched from $[0, 1]$ to $[-1, 1]$.
- **Why it matters:** It gives the “perfectly spread corpus” reference curve in closed form: no simulation of fake corpora is needed (one draws random *numbers* rather than random *vectors*), which is what makes 199-replicate null bands cheap.
- **Term by term:**
 - c — the pair cosine whose null distribution is being stated.

- B — a random draw from the Beta distribution, which produces numbers between 0 and 1.
- $\text{Beta}(a, a)$ with $a = (d - 1)/2$ — a two-parameter family of distributions on $[0, 1]$; with both parameters equal it is a symmetric bell around $1/2$ that narrows as dimension grows (the concentration phenomenon again).
- $2B - 1$ — rescales $[0, 1]$ onto $[-1, 1]$ so the result is a cosine: $B = 1/2$ (the typical draw) maps to $c = 0$, i.e. perpendicular.

The anisotropy-matched reference (ACG): $x = \sqrt{\lambda} \odot g / \|\sqrt{\lambda} \odot g\|$.

- **What it says:** A second reference corpus that copies the real corpus’s overall *shape* — its elongation along some directions and flattening along others — but contains no clustering whatsoever.
- **Why it matters:** Learned embeddings almost always occupy a narrow cone (the “cone effect”), which inflates all cosines benignly. Comparing the data against this reference, instead of against the uniform null alone, separates crowding the cone explains (benign) from crowding it cannot (genuine pockets).
- **Term by term:**
 - g — a vector of d independent standard Gaussian random numbers: a direction chosen with no preference at all.
 - λ — the corpus’s covariance eigenvalues: a list of d non-negative numbers, one per principal direction, each saying how much variance the corpus has along that direction. Together they describe the corpus’s ellipsoidal shape.
 - $\sqrt{\lambda} \odot g$ — multiply each coordinate of g by the square root of the matching eigenvalue ($\odot$ is coordinate-wise multiplication); this stretches the isotropic random cloud into the corpus’s ellipsoid.
 - dividing by $\|\cdot\|$ — rescale to length one, so the reference point lands on the sphere like the real data.
 - Only the eigenvalues enter — rotating a corpus changes none of its pairwise cosines, so the *eigenvectors* are irrelevant here.

The global rank-envelope test and the liftoff cosine.

- **What it says:** The test decides whether the data curve $K(t)$ exceeds the null band across *all* thresholds at once, rather than at one chosen threshold. It simulates many null curves, ranks the data curve among them at every threshold simultaneously, and returns one global p -value. The *liftoff cosine* is the smallest similarity at which the data curve exits the envelope, reported only when the global test rejects.
- **Why it matters:** Reading exceedance off a pointwise band at each threshold separately commits a multiple-comparison error — on pure noise it fires essentially always (measured: every one of 40 null corpora). The global test’s false alarms match its nominal rate.

- **How it is used:** Every liftoff claim in the report is gated on this global p ; extreme-rank-length (ERL) ordering breaks ties among simulated curves.
- **Term by term:**
 - envelope — the band spanned by the simulated null curves (here 199 of them, drawn via the Beta law above).
 - p -value — the probability, if the corpus were truly well-spread, of seeing a curve as extreme as the data’s anywhere along its length. Small p (e.g. 0.005) means “a well-spread corpus essentially never looks like this.” With 199 simulations the smallest attainable p is $1/200 = 0.005$.
 - ERL (extreme rank length) — the tie-breaking rule: when two curves are equally extreme at their worst point, the one that stays extreme over a longer stretch ranks as more extreme.
 - liftoff cosine — the onset similarity of genuine tight-pair structure: below it the data track the reference; above it they exceed the envelope.

Stolarsky invariance and the occupancy discrepancy z : cap discrepancy = $\text{const} - \frac{1}{n^2} \sum_{i,j} \|x_i - x_j\|$.

- **What it says:** Take every possible “cap” on the sphere (every center, every radius), compare how many points each cap holds against how many it should hold, square the errors, and add them all up. Stolarsky’s theorem says this total — which looks uncomputable — equals a constant minus the average distance between pairs of points.
- **Why it matters:** It is the exact completion of what binned displays approximate: *every* lattice at *every* scale simultaneously, with no lattice to choose. A corpus that under-spreads (crowds anywhere, at any scale) has a smaller average pairwise distance.
- **How it is used:** The report computes the mean pairwise chord and standardizes it against the uniform null, giving the occupancy discrepancy z . On the benchmark, uniform controls score $z \approx 0$ and the planted pocket scores $z = -47$.
- **Term by term:**
 - cap discrepancy — the total squared error between actual and ideal cap occupancy, accumulated over all cap centers and radii; the formal object on the left of the identity.
 - const — an explicit constant that depends only on n and d , not on where the points are; it plays no role in comparisons.
 - $\frac{1}{n^2} \sum_{i,j} \|x_i - x_j\|$ — the average chord distance over all ordered pairs of documents (sum over every pair, divide by the number of pairs).
 - z — the reported score: the observed mean chord minus the null mean, divided by the null standard deviation, where the null mean and standard deviation come from redrawing the same number of pair cosines from the Beta law. $z = -47$ reads “the corpus’s average distance sits 47 standard deviations of sampling noise below the well-spread value.”

Distance to a measure: $\text{DTM}_m(x_i) = \left(\frac{1}{k} \sum_{j=1}^k r_{i,(j)}^2\right)^{1/2}$ **with** $k = \lceil mn \rceil$.

- **What it says:** For each item: the typical distance it must reach to gather the nearest m -fraction of the corpus around it — a per-item crowding radius. Small radius = crowded; large radius = isolated.
- **Why it matters:** It moves from corpus-level statistics to *named items*, which is what makes findings actionable. It is also provably stable: outliers and small perturbations of the corpus move it only slightly (the Wasserstein-stability guarantee).
- **How it is used:** Computed for every reservoir item; its low tail (the most crowded items) is compared against the uniform corpus’s band; the report lists the most-crowded and most-isolated items by identifier.
- **Term by term:**
 - $\text{DTM}_m(x_i)$ — the function’s name (“distance to the measure”) applied to document i at mass fraction m ; one number per document.
 - m — the mass fraction, the method’s one knob (e.g. 0.02: gather 2% of the corpus); being a fraction makes it scale-free in corpus size.
 - $k = \lceil mn \rceil$ — that fraction converted to a count of neighbors; $\lceil \cdot \rceil$ rounds up to a whole number.
 - $r_{i,(j)}$ — the distance from item i to its j th nearest neighbor; the parenthesized subscript (j) means the distances have been sorted ascending, so (1) is the closest.
 - $\frac{1}{k} \sum_{j=1}^k r_{i,(j)}^2$ — the average of the squared distances to the k nearest neighbors.
 - $(\dots)^{1/2}$ — the square root of that average: a root-mean-square radius, an average robust to any single neighbor.

The merge tree: mutual reachability, birth, death, prominence.

- **What it says:** Connect items by their shortest bridges (single linkage) under the *mutual-reachability* distance. As the connection scale grows, tight groups form and later dissolve into the bulk. A pocket’s *birth* is the scale at which it first holds together, its *death* the scale at which it merges into something larger, and its *prominence* = death – birth is how long it exists as its own entity.
- **Why it matters:** Prominence replaces every flat clustering threshold: a pocket is reported not because it passed a cutoff but because it persists across a wide range of scales. A null-calibrated prominence floor (the 99th percentile of prominence produced by pure noise at matched size and dimension) suppresses spurious pockets.
- **Term by term:**
 - single linkage — build connections by repeatedly merging the two closest groups; the merge tree records the order and scale of every merge.

- mutual reachability of a, b — $\max\{d(a, b), \text{core}(a), \text{core}(b)\}$: the largest of the three listed quantities, where $d(a, b)$ is the ordinary distance between the two points and $\text{core}(x)$ is the distance from x to its own k th nearest neighbor. Taking the max means two points count as close only if each is also locally dense — a noise point far from everything cannot form a spurious bridge.
- birth / death — the connection scales at which a group first assembles and at which it is absorbed into an older group (the *elder rule*: when two groups meet, the younger one dies).
- prominence — death minus birth, in distance units: the width of the scale range over which the pocket is a separate entity.
- excess of mass — the selection rule choosing, on each branch of the tree, the cluster that persists longest.

The query-noise model: $q = x + \sigma g$, $g \sim \mathcal{N}(0, I_d)$.

- **What it says:** A query aimed at item x lands nearby rather than exactly on it: x plus Gaussian noise of scale σ in every direction.
- **Why it matters:** It converts geometry into operational risk. Real queries paraphrase, truncate, and typo; σ is the abstract knob for all such imprecision, and every downstream quantity is a function of it.
- **Term by term:**
 - q — the query vector the system actually searches with.
 - x — the item the query “means”: its intended target.
 - $g \sim \mathcal{N}(0, I_d)$ — a fresh standard Gaussian vector: each of the d coordinates is an independent bell-curve draw with mean 0 and variance 1 (I_d , the identity matrix, encodes “independent, unit variance, no preferred direction”).
 - σ — the noise scale: how far, typically, the query lands from its target, in the same units as chord distance. $\sigma = 0$ means perfect queries; larger σ means sloppier ones.

The exact flip law: $\Pr[\langle q, y \rangle > \langle q, x \rangle] = \Phi(-r/2\sigma)$.

- **What it says:** The probability that a noisy query aimed at x scores a competitor y higher than x itself depends on nothing but the distance between x and y , through one evaluation of the normal tail. Exactly — in any dimension.
- **Why it matters:** It is the bridge between geometry and retrieval errors, and it explains why near-duplicates are catastrophic: as $r \rightarrow 0$ the probability approaches $\Phi(0) = 1/2$ — a duplicated document loses half its queries to its twin no matter how small the noise is.
- **How it is used:** Verified by direct simulation to Monte-Carlo precision (max deviation 9×10^{-4}); summed over competitors to give $C(\sigma)$; used as the training loss kernel.
- **Term by term:**

- x, y — the query’s intended target and one competing document, both unit vectors;
 q — the noisy query from the previous entry.
- $\langle q, y \rangle$ — the dot product of query and competitor: exactly the similarity score retrieval ranks by. The bracketed statement $\langle q, y \rangle > \langle q, x \rangle$ reads “the competitor scores higher than the intended target,” i.e. the retrieval errs.
- $\Pr[\dots]$ — the probability of that error over the randomness of the query noise g .
- r — the chord distance between x and y ; σ — the query-noise scale.
- $\Phi(-r/2\sigma)$ — the normal tail evaluated at $-r/2\sigma$. Why this argument: the set of points scoring x and y equally is the perpendicular bisector of the segment joining them, which sits at distance $r/2$ from x ; Gaussian noise of scale σ crosses a boundary $r/2$ away with probability $\Phi(-(r/2)/\sigma)$. Larger separation or smaller noise pushes the argument more negative and the error probability toward zero.

Expected collisions: $C(\sigma) = \sum_{j \neq i} \Phi(-r_{ij}/2\sigma) \approx (n-1) \int \Phi(-r/2\sigma) dK(r)$.

- **What it says:** For an average item, the expected number of wrong competitors that a noisy query would rank above it: add the flip probability over every competitor.
- **Why it matters:** The probability that *any* of several events occurs is at most the sum of their individual probabilities (Boole’s inequality). $C(\sigma)$ therefore bounds the probability of *any* retrieval error, however the competitors are correlated. The bound errs only on the safe side: crowding can make it loose, never wrong.
- **Term by term:**
 - $C(\sigma)$ — the function’s name: expected collision count at noise scale σ ; one number per σ , increasing in σ .
 - i — the item being queried; j — each competing document in turn.
 - $\sum_{j \neq i} \Phi(-r_{ij}/2\sigma)$ — the flip probability against competitor j (previous entry), added up over all $n - 1$ competitors.
 - $(n - 1) \int \Phi(-r/2\sigma) dK(r)$ — the same sum computed from the pair-distance distribution alone: average the flip probability over the distribution of pair distances (that is the integral against dK), then multiply by the number of competitors. This is why a random pair *sample* suffices — no per-item computation is needed for the corpus-level curve.

Resolution bandwidth: $\sigma^* = \max\{\sigma : C(\sigma) \leq 1\}$.

- **What it says:** The largest amount of query noise the corpus can absorb before, in expectation, one wrong item outranks the right one.
- **Why it matters:** It is the report’s headline operational scalar: a noise *budget* in retrieval units. Comparing σ^* against the same corpus size spread uniformly (the ratio reported alongside it) says what fraction of the theoretically available budget the embedding retains.

- **Term by term:**

- $\{\sigma : C(\sigma) \leq 1\}$ — the set of all noise scales at which the expected collision count stays at or below one; max takes the largest such scale.
- the threshold 1 — “one expected collision per item”: the point where retrieval starts, on average, to lose. It is a convention (a tolerance), and the report states it wherever σ^* appears.
- C is monotone — more noise always means more collisions — so this largest admissible σ is well defined and is found by bisection: repeatedly halve an interval known to contain the crossing.

Uniformity as a collision bound: $U_t = \log \mathbb{E} e^{-t\|x-y\|^2}$ and $C(\sigma) \leq (n-1) e^{U_t}$ at $t = 1/(8\sigma^2)$.

- **What it says:** The widely used Wang–Isola “uniformity” loss — the log of the average of e^{-tr^2} over pairs — is, at the right temperature, exactly the logarithm of a Chernoff upper bound on the collision count $C(\sigma)$.
- **Why it matters:** It gives the popular loss operational units: its temperature knob t is not folklore but a choice of query-noise scale ($t = 2$ corresponds to $\sigma = 0.25$). It also explains when uniformity misleads — it is a bound, and the report measures where the bound’s ordering and the exact statistic’s ordering disagree.

- **Term by term:**

- U_t — the uniformity statistic’s value at temperature t (one number per t ; more negative = more spread).
- x, y — a random pair of documents; $\mathbb{E}$ — the average over that random pair.
- $e^{-t\|x-y\|^2}$ — a Gaussian-shaped penalty: 1 when the pair coincides, decaying with squared distance at rate t (large t = only very close pairs are penalized).
- $\log$ — the natural logarithm of the average, following the original definition.
- $t = 1/(8\sigma^2)$ — the dictionary between temperature and noise scale; where the identity holds.
- the proof, in one line — the normal tail bound $\Phi(-z) \leq e^{-z^2/2}$ (a Chernoff bound) applied to $z = r/2\sigma$ turns each flip probability into $e^{-r^2/8\sigma^2}$, and summing gives $(n-1) e^{U_t}$.

The global suite: mean pair cosine, IsoScore, effective rank, hubness skew.

- **What it says:** Four whole-corpus summaries used as context and as baselines. Mean pair cosine: the average c_{ij} over random pairs (how coned the corpus is). IsoScore: how close the covariance is to perfectly isotropic — equal variance in all directions — on a 0-to-1 scale. Effective rank: exp of the entropy of the normalized eigenvalue distribution — roughly, how many directions carry meaningful variance (between 1 and d). Hubness skew: the skewness (asymmetry) of the count of how often each item appears in others’ k -nearest-neighbor lists — heavy right skew means a few “hub” items intrude into everyone’s neighborhoods.

- **Why it matters:** These are the statistics that practitioners already consult; the detection-power study shows them blind to small pockets that the continuous layer detects at full power, because they are global averages.

Training losses: $\mathcal{L}_{\text{conf}} = \text{mean}_{(i,j) \notin \mathcal{E}} \Phi(-\|z_i - z_j\|/2\sigma)$ and $\mathcal{L}_{\text{pres}} = \text{mean}_i \text{KL}(p_i^{\text{base}} \| p_i)$.

- **What it says:** Two differentiable objectives built from the measurements. The first is the batch version of the collision count: the average flip probability over batch pairs, at the *measured* noise scale, excluding a set of protected pairs. The second pins local semantics: for each anchor, the distribution “who in this batch is similar to me” under the model being trained must stay close to the same distribution under the frozen base model.
- **Why it matters:** $\mathcal{L}_{\text{conf}}$ spends gradient only where confusion actually lives, and $\mathcal{L}_{\text{pres}}$ prevents the repair from destroying meaning. Their balance (λ_p) is the loop’s central tension: recovery against fidelity.
- **Term by term:**
 - $\mathcal{L}$ — a loss: a number training tries to make small; the subscripts name the two losses (confusion, preservation).
 - z_i — the embedding of batch item i under the model currently being trained (it changes every step; x_i is reserved for the fixed corpus vectors).
 - $\text{mean}_{(i,j) \notin \mathcal{E}}$ — average over all ordered batch pairs *except* those in the excluded set; $\notin$ means “not a member of.”
 - $\mathcal{E}$ — the excluded pairs: known positives and the false-negative guard (each anchor’s top- m base-model neighbors, which are likely unlabeled true relatives and must not be pushed apart).
 - $\Phi(-\|z_i - z_j\|/2\sigma)$ — the flip law applied to the batch pair at the measured scale $\sigma = \sigma^*$.
 - $\text{KL}(p \| q)$ — the Kullback–Leibler divergence: the standard measure of how much a distribution q misrepresents a reference distribution p ; zero exactly when they agree, larger the more they differ. The double bar $\|$ is part of the notation and separates reference from candidate.
 - p_i and p_i^{base} — for anchor i , the softmax over its cosines to the rest of the batch (exponentiate each cosine divided by a temperature of 0.05, then normalize to sum to one — low temperature sharpens the distribution toward the nearest neighbors), computed under the trained model and under the frozen base model respectively.

Gradient locality: $\partial\Phi(-r/2\sigma)/\partial r \propto e^{-r^2/8\sigma^2}$.

- **What it says:** The derivative of the flip probability with respect to pair distance dies off as a Gaussian in r : pairs farther apart than about 3σ receive exponentially negligible gradient.

- **Why it matters:** It is a structural safety guarantee: minimizing $\mathcal{L}_{\text{conf}}$ *cannot* disturb the corpus’s far-pair (dissimilarity) structure, by construction rather than by tuning. The test suite enforces it (a near pair receives more than 10^3 times the gradient of an orthogonal pair).
- **Term by term:**
 - $\partial\Phi(-r/2\sigma)/\partial r$ — how fast the flip probability changes as the pair distance r changes: exactly the signal backpropagation sends to a pair during training.
 - $\propto$ — proportional to: equal up to a constant that does not involve r .
 - $e^{-r^2/8\sigma^2}$ — a Gaussian in r : near 1 for close pairs, and collapsing extremely fast once r exceeds a few multiples of σ — at $r = 3\sigma$ it is already below $e^{-9/8} \approx 0.32$ of its peak and falling as the *square* of the distance in the exponent.

Neighbor overlap at k .

- **What it says:** For each item, the fraction of its k nearest neighbors that two embeddings (base and tuned) have in common; the report uses $k = 10$ and averages over held-out items.
- **Why it matters:** It is the retrieval-relevant measure of “did the model change who is similar to whom” — comparing neighbor *identities*, which global similarity statistics can miss — and it is the loop’s licensing criterion (≥ 0.90).
- **Term by term:**
 - k — the neighborhood size: how many nearest neighbors each item’s list contains (10 here, matching a retrieval page).
 - overlap for one item — $|A \cap B|/k$, where A and B are the item’s neighbor sets under the two embeddings, $\cap$ is the set intersection (members common to both), and $|\cdot|$ counts members: 1 means identical neighborhoods, 0 fully reshuffled.

The statistical tests used in this report.

- **McNemar’s exact test** asks whether a model change helped or hurt on *paired* outcomes: it looks only at the discordant queries (helped by one model, failed by the other — here 118 against 131) and asks whether that split is farther from an even division than chance alone would produce.
- **The paired sign test** does the same for per-item quantities: of 408 held-out flagged items, 406 improved — the probability of a split that uneven under “no effect” is vanishingly small ($p \approx 10^{-118}$), which is how a tiny median effect can still be unambiguous.
- **Bootstrap confidence intervals** quantify uncertainty without distributional assumptions: resample the data with replacement many times, recompute the statistic each time, and read the spread; an interval containing zero means the sign of the effect is not established.

- **AUC** (area under the ROC curve, equivalently the Mann–Whitney statistic) measures a predictor’s ability to rank: the probability that a randomly chosen failing document scores higher on the predictor than a randomly chosen non-failing one — 0.5 is chance, 1.0 is perfect ordering.

References

- [1] J. Gao, D. He, X. Tan, T. Qin, L. Wang, T.-Y. Liu. Representation degeneration problem in training natural language generation models. *ICLR*, 2019.
- [2] K. Ethayarajh. How contextual are contextualized word representations? *EMNLP*, 2019.
- [3] J. Mu, S. Bhat, P. Viswanath. All-but-the-top: simple and effective postprocessing for word representations. *ICLR*, 2018.
- [4] X. Cai, J. Huang, Y. Bian, K. Church. Isotropy in the contextual embedding space: clusters and manifolds. *ICLR*, 2021.
- [5] O. Roy, M. Vetterli. The effective rank: a measure of effective dimensionality. *EUSIPCO*, 2007.
- [6] W. Rudman, N. Gillman, T. Rush, C. Eickhoff. IsoScore: measuring the uniformity of embedding space utilization. *Findings of ACL*, 2022.
- [7] T. Wang, P. Isola. Understanding contrastive representation learning through alignment and uniformity on the hypersphere. *ICML*, 2020.
- [8] M. Radovanović, A. Nanopoulos, M. Ivanović. Hubs in space: popular nearest neighbors in high-dimensional data. *JMLR* 11, 2010.
- [9] Q. Garrido, R. Balestrieri, L. Najman, Y. LeCun. RankMe: assessing the downstream performance of pretrained self-supervised representations by their rank. *ICML*, 2023.
- [10] K. Agrawal, A. Mondal, A. Ghosh, B. Richards. α -ReQ: assessing representation quality by measuring eigenspectrum decay. *NeurIPS*, 2022.
- [11] A. Tsitsulin, M. Munkhoeva, B. Perozzi. Unsupervised embedding quality evaluation. *TAG-ML @ ICML*, 2023.
- [12] D. Friedman, A. B. Dieng. The Vendi score: a diversity evaluation metric for machine learning. *TMLR*, 2023.
- [13] A. Pasarkar, A. B. Dieng. Cousins of the Vendi score: a family of similarity-based diversity metrics. *AISTATS*, 2024.
- [14] L. G. S. Giraldo, M. Rao, J. C. Príncipe. Measures of entropy from data using infinitely divisible kernels. *IEEE Trans. Information Theory* 61(1), 2015.
- [15] A. Ospanov, F. Farnia. Do Vendi scores converge with finite samples? Truncated Vendi score for finite-sample convergence guarantees. arXiv:2410.21719, 2024.
- [16] B. D. Ripley. The second-order analysis of stationary point processes. *J. Applied Probability* 13(2), 1976.
- [17] J. Besag. Contribution to the discussion of Dr Ripley’s paper. *JRSS-B* 39(2), 1977.

- [18] S. M. Robeson, A. Li, C. Huang. Point-pattern analysis on the sphere. *Spatial Statistics* 10, 2014.
- [19] A. J. Baddeley, J. Møller, R. Waagepetersen. Non- and semi-parametric estimation of interaction in inhomogeneous point patterns. *Statistica Neerlandica* 54(3), 2000.
- [20] D. E. Tyler. Statistical analysis for the angular central Gaussian distribution on the sphere. *Biometrika* 74(3), 1987.
- [21] M. Myllymäki, T. Mrkvička, P. Grabarnik, H. Seijo, U. Hahn. Global envelope tests for spatial processes. *JRSS-B* 79(2), 2017.
- [22] K. B. Stolarsky. Sums of distances between points on a sphere II. *Proc. AMS* 41(2), 1973.
- [23] J. S. Brauchart, J. Dick. A simple proof of Stolarsky’s invariance principle. *Proc. AMS* 141, 2013 (arXiv:1101.4448).
- [24] C. E. Gehlke, K. Biehl. Certain effects of grouping upon the size of the correlation coefficient in census tract material. *JASA* 29(185A), 1934.
- [25] S. Openshaw, P. J. Taylor. A million or so correlation coefficients: three experiments on the modifiable areal unit problem. In *Statistical Applications in the Spatial Sciences*, Pion, 1979.
- [26] S. Openshaw. *The Modifiable Areal Unit Problem*. CATMOG 38, Geo Books, 1984.
- [27] W. S. Robinson. Ecological correlations and the behavior of individuals. *American Sociological Review* 15(3), 1950.
- [28] D. W. Scott. On optimal and data-based histograms. *Biometrika* 66(3), 1979.
- [29] D. Freedman, P. Diaconis. On the histogram as a density estimator: L_2 theory. *Z. Wahrsch. Verw. Gebiete* 57, 1981.
- [30] D. W. Scott. Averaged shifted histograms: effective nonparametric density estimators in several dimensions. *Annals of Statistics* 13(3), 1985.
- [31] B. W. Silverman. *Density Estimation for Statistics and Data Analysis*. Chapman & Hall, 1986.
- [32] F. Chazal, D. Cohen-Steiner, Q. Mérigot. Geometric inference for probability measures. *Found. Comput. Math.* 11(6), 2011.
- [33] F. Chazal, B. Michel. An introduction to topological data analysis: fundamental and practical aspects for data scientists. *Frontiers in AI* 4, 2021.
- [34] F. Chazal, L. J. Guibas, S. Y. Oudot, P. Skraba. Persistence-based clustering in Riemannian manifolds. *J. ACM* 60(6), 2013.
- [35] R. J. G. B. Campello, D. Moulavi, A. Zimek, J. Sander. Hierarchical density estimates for data clustering, visualization, and outlier detection. *ACM TKDD* 10(1), 2015.
- [36] A. Rolle, L. Scoccola. Stable and consistent density-based clustering via multiparameter persistence. *JMLR*, 2024 (arXiv:2005.09048).
- [37] J. A. Hartigan. Consistency of single linkage for high-density clusters. *JASA* 76(374), 1981.
- [38] K. Chaudhuri, S. Dasgupta. Rates of convergence for the cluster tree. *NeurIPS*, 2010.
- [39] D. Cohen-Steiner, H. Edelsbrunner, J. Harer. Stability of persistence diagrams. *Discrete Comput. Geom.* 37, 2007.

- [40] K. Beyer, J. Goldstein, R. Ramakrishnan, U. Shaft. When is “nearest neighbor” meaningful? *ICDT*, 1999.
- [41] E. Levina, P. J. Bickel. Maximum likelihood estimation of intrinsic dimension. *NeurIPS*, 2005.
- [42] E. Facco, M. d’Errico, A. Rodriguez, A. Laio. Estimating the intrinsic dimension of datasets by a minimal neighborhood information. *Scientific Reports* 7, 2017.
- [43] M. Wattenberg, F. Viégas, I. Johnson. How to use t-SNE effectively. *Distill*, 2016.
- [44] A. Narayan, B. Berger, H. Cho. Assessing single-cell transcriptomic variability through density-preserving data visualization. *Nature Biotechnology* 39, 2021.
- [45] S. Kornblith, M. Norouzi, H. Lee, G. Hinton. Similarity of neural network representations revisited. *ICML*, 2019.
- [46] V. Pappayan, X. Y. Han, D. L. Donoho. Prevalence of neural collapse during the terminal phase of deep learning training. *PNAS* 117(40), 2020.
- [47] Qwen Team. Qwen3 Embedding: advancing text embedding and reranking through foundation models. *arXiv:2506.05176*, 2025.
- [48] S. Iyer, N. Dandekar, K. Csernai. First Quora dataset release: question pairs. Quora Engineering Blog, 2017.
- [49] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, M. Zhou. MiniLM: deep self-attention distillation for task-agnostic compression of pre-trained transformers. *NeurIPS*, 2020.
- [50] K. Song, X. Tan, T. Qin, J. Lu, T.-Y. Liu. MPNet: masked and permuted pre-training for language understanding. *NeurIPS*, 2020.
- [51] N. Thakur, N. Reimers, A. Rüklé, A. Srivastava, I. Gurevych. BEIR: a heterogeneous benchmark for zero-shot evaluation of information retrieval models. *NeurIPS Datasets and Benchmarks*, 2021.
- [52] E. Voorhees, T. Alam, S. Bedrick, D. Demner-Fushman, W. R. Hersh, K. Lo, K. Roberts, I. Soboroff, L. L. Wang. TREC-COVID: constructing a pandemic information retrieval test collection. *SIGIR Forum* 54(1), 2020.
- [53] Z. Yang, P. Qi, S. Zhang, Y. Bengio, W. W. Cohen, R. Salakhutdinov, C. D. Manning. HotpotQA: a dataset for diverse, explainable multi-hop question answering. *EMNLP*, 2018.
- [54] T. Kwiatkowski et al. Natural Questions: a benchmark for question answering research. *TACL* 7, 2019.
- [55] C. K. Reddy, L. Márquez, F. Valero, N. Rao, H. Zaragoza, S. Bandyopadhyay, A. Biswas, A. Xing, K. Subbian. Shopping queries dataset: a large-scale ESCI benchmark for improving product search. *arXiv:2206.06588*, 2022.
- [56] N. Reimers, I. Gurevych. Sentence-BERT: sentence embeddings using Siamese BERT-networks. *EMNLP*, 2019.
- [57] L. Wang, N. Yang, X. Huang, B. Jiao, L. Yang, D. Jiang, R. Majumder, F. Wei. Text embeddings by weakly-supervised contrastive pre-training. *arXiv:2212.03533*, 2022.
- [58] S. Xiao, Z. Liu, P. Zhang, N. Muennighoff. C-Pack: packaged resources to advance general Chinese and English text embedding. *SIGIR*, 2024.

- [59] L. Merrick, D. Xu, G. Nuti, D. Campos. Arctic-Embed: scalable, efficient, and accurate text embedding models. *arXiv:2405.05374*, 2024.
- [60] Gemma Team. EmbeddingGemma: efficient open text embeddings (model release and technical documentation), 2025.
- [61] NVIDIA. Nemotron-Pretraining-Legal-v1 (dataset, CC-BY-4.0), 2026. <https://huggingface.co/datasets/nvidia/Nemotron-Pretraining-Legal-v1>; the dataset card directs citation to the NVIDIA Nemotron 3 Ultra technical report.